

Goal-directed remembering under contextual competition: A retrieved-context account of selective interference

Jordan B Gunn

Department of Psychology, University of Cambridge

Author Note

This is an in-progress working draft. Please do not cite, quote, or circulate without the authors' written permission.

Abstract

Goal-directed remembering can be less affected by interference even as the same material becomes less likely to come to mind inadvertently. The trauma-film paradigm examines this asymmetry in memory for distressing films: a visuospatial intervention task performed after a film reminder can reduce later intrusions while leaving voluntary memory relatively intact. This selective interference effect is often interpreted through theories of traumatic memory in which intrusive and deliberate remembering depend on differently vulnerable memory representations. Here we show that the same dissociation may follow from retrieved-context dynamics developed to explain episodic memory more generally. We develop this account in an emotional Context Maintenance and Retrieval (eCMR) model that predicts sequences of film and non-film recall events. The model links task items to film-associated context reinstated by a film reminder. That context can later cue task competitors as well as film items. Unguided film recall declines when reminder-linked learning makes task items strong competitors. Deliberate recall is less affected when a maintained film-category cue favors film candidates without equally supporting task competitors. Recognition is less affected than recall because the model evaluates a presented candidate item's fit to context rather than selecting among film and task candidates. This account clarifies when intrusive and voluntary remembering can diverge and situates selective interference within a broader computational theory of episodic learning and retrieval.

Keywords: intrusive memories, voluntary memory, trauma-film paradigm, visuospatial intervention, selective interference, retrieved-context theory, computational modeling

Goal-directed remembering under contextual competition: A retrieved-context account of selective interference

Introduction

Traumatic experience foregrounds a tension between accessibility and selectivity in episodic memory. From a functional perspective, events with lasting significance need to remain accessible in memory so remembered details can guide later decisions (Zhou et al., 2025). At the same time, adaptive retrieval is selective: what comes to mind should normally depend on current goals, cues, and circumstances. Intrusive memories after trauma show how highly accessible event memories can strain the selectivity of retrieval. In PTSD, such memories can come to mind unbidden and poorly matched to the present, causing distress and interfering with ongoing activity (Brewin et al., 2010; Ehlers & Clark, 2000; Iyadurai et al., 2019; Krans et al., 2009). Explaining how episodic memory keeps consequential event memories accessible while regulating their retrieval is therefore a shared target for both clinical science and mechanistic theories of memory.

The trauma-film paradigm provides a controlled setting for comparing intrusive and voluntary memory for the same event (Holmes & Bourne, 2008; James et al., 2016). Participants view distressing film material, record later intrusive memories of the film, and often complete voluntary-memory tests for the same material. This design makes it possible to ask whether the same encoded event can remain accessible to deliberate retrieval while becoming more or less likely to come to mind involuntarily. In such research, intrusion frequency is often only weakly related to voluntary-memory performance, such that the two measures dissociate (Holmes et al., 2004; James et al., 2016). The same encoded episode can therefore support both involuntary and deliberate remembering, and these modes can come apart empirically.

Visuospatial interference tasks make this dissociation experimentally sharper. In immediate post-film studies, tasks such as Tetris can reduce later film intrusions while voluntary memory for the same material remains relatively intact (Deeprrose et al., 2012; Holmes et al., 2009, 2010; Lau-Zhu et al., 2019). Delayed variants pair the interference task with a film reminder. Participants return after roughly 24 hours to 3 days, view reminder cues for the film,

and then complete Tetris. This reminder-plus-Tetris sequence can reduce subsequent intrusions (James et al., 2015; Kessler et al., 2020). The resulting pattern of reduced intrusions with relatively preserved deliberate memory is termed selective interference. Figure 1 defines the qualitative pattern targeted by the recall simulations: following a film reminder, the visuospatial task has a larger effect on unguided than deliberate film recall.

The selective interference effect has been used to argue that intrusive and voluntary remembering depend on differently vulnerable memory representations (Brewin et al., 1996; Brewin, 2014). Dual-representation accounts give this interpretation an architectural basis. They link coherent deliberate recollection to hippocampally supported representations that bind event elements within a spatiotemporal context (Bird et al., 2012; Bisby et al., 2020). Intrusive re-experiencing is linked more strongly to sensory and affective representations supported by amygdala and sensory systems (Bisby et al., 2020; Brewin et al., 2010). In immediate post-film interventions, visuospatial tasks are understood to compete for working-memory resources while sensory-perceptual imagery from the film is still being maintained and consolidated (Baddeley & Andrade, 2000). Because later intrusions are assumed to arise especially from sensory-bound representations, the selective pattern is attributed to disrupting the representation most likely to support intrusions while leaving voluntary-memory performance relatively preserved (Deepröse et al., 2012; Holmes et al., 2009, 2010; Lau-Zhu et al., 2019). In delayed post-reminder interventions, the reminder is instead treated as reactivating the relevant memory before the task, allowing the interference procedure to alter later availability through reconsolidation or post-reactivation updating (James et al., 2015; Kindt et al., 2009). Across these variants, selective interference is explained by treating the intervention as acting more strongly on sensory-bound, intrusion-supporting representations than on the contextualized representations proposed to support coherent deliberate recollection.

Here we argue that preserved voluntary memory does not, by itself, establish that a separate representation was spared. Selective interference can instead arise from differences in how film memories are cued and controlled at retrieval. In the account developed below, a film

reminder reinstates context associated with the film before task material is encoded. Task material then becomes linked to that context. When later remembering is guided by film-associated context, task and film material can therefore compete, interfering with film recall. The task interferes less when retrieval depends less on context to access film material. Deliberate recall can be less affected when a maintained film-category cue gives film candidates an additional source of support without equally supporting task competitors. Tasks like recognition can also be less affected, because the test presents a candidate film item for judgment rather than requiring film material to be brought to mind. On this account, the selective-interference effect emerges when retrieval conditions differ in how much they depend on film-associated context that also cues task material.

We formalize this retrieval account in an emotional variant ([Cohen & Kahana, 2022](#); [Talmi et al., 2019](#)) of the Context Maintenance and Retrieval (CMR) model ([Polyn et al., 2009a](#)). Retrieved-context models treat context as a gradually changing state that is shaped by experience, reinstated by retrieval, and used to cue later memory search ([Howard & Kahana, 2002](#); [Polyn et al., 2009a](#); [Sederberg et al., 2008](#)). In CMR, items and context are linked by bidirectional associations. Context cues candidate items, and retrieved items or items presented at test reinstate associated context ([Polyn et al., 2009a, 2009b](#)). Emotional CMR variants retain this item-context architecture while allowing emotional or source context to shape what is bound and later retrieved ([Cohen & Kahana, 2022](#); [Talmi et al., 2019](#)). These properties make CMR a principled setting for a retrieved-context account of selective interference based on context reinstatement and item-context binding.

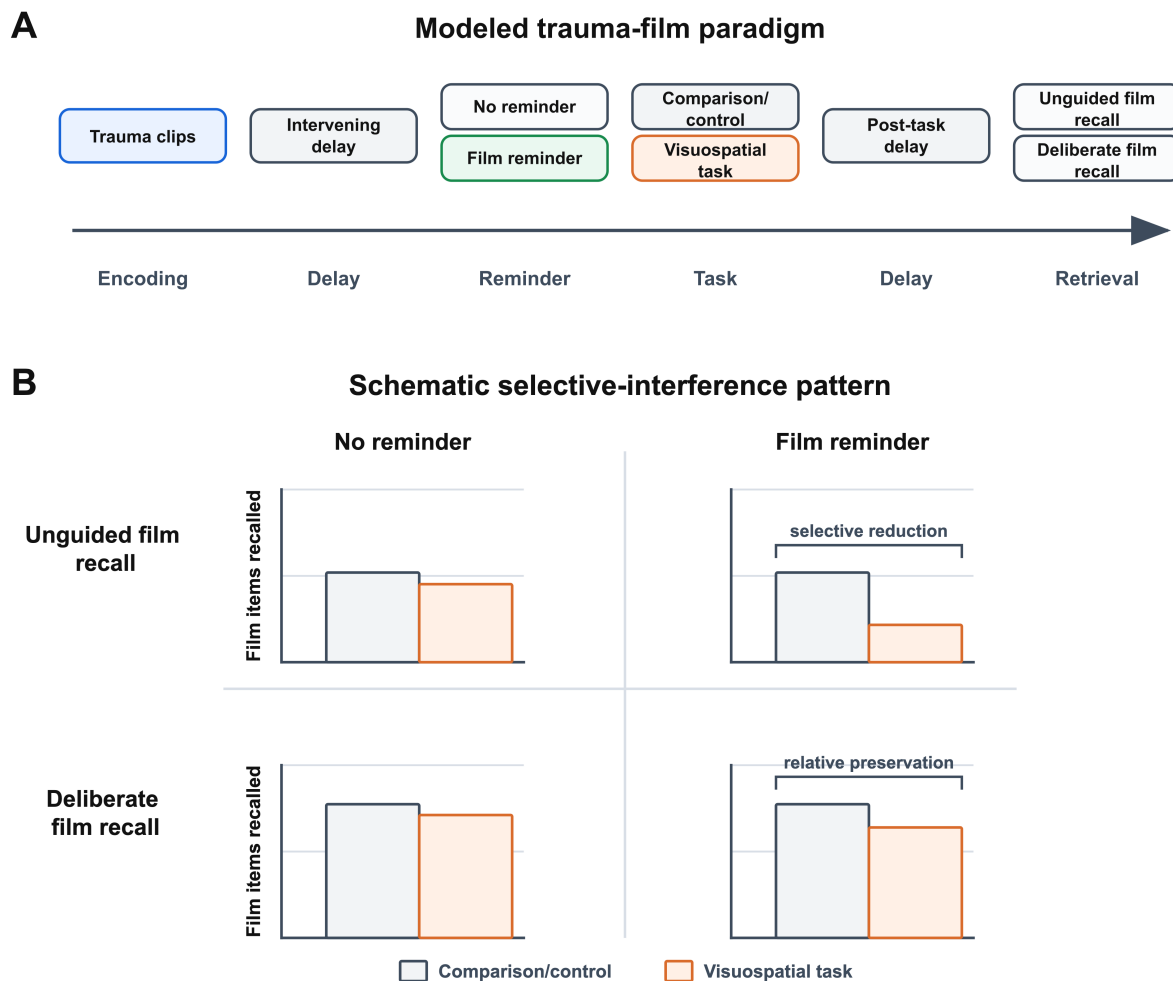
Applied to the trauma-film paradigm, the model treats film, delay-period, task, and filler events as a continuous sequence through which context evolves. A film reminder is implemented as item-to-context reinstatement before task encoding. Later recall is implemented as context-to-item sampling from the film and non-film items supported by the current retrieval state. When that context supports both film and task material, task items compete with film recall rather than merely weakening a film trace.

Deliberate recall is modeled with several retrieval-control processes. Rather than universally producing the selective-interference pattern, we show that some control mechanisms increase film recall overall without substantially reducing the loss caused by task competitors linked to film-associated context. In the simulations, the strongest protection comes from adding a maintained film-category cue that gives film candidates support not shared by task competitors. This identifies a functional condition for selective interference: voluntary recall is protected when it depends less on film-associated temporal context that has also become supportive of task material.

The simulations follow a generalization-style modeling logic ([Busemeyer & Wang, 2000](#); [Lohnas et al., 2023](#)). Core parameters are anchored in an independent affective free-recall dataset ([Dupertuys et al., 2026](#)), and no trauma-film outcomes are fitted. Simulation 1 varies reminder presence and effective task-context learning to test competitor formation. Simulation 2 compares start-of-film context reinstatement, a maintained film-category cue, and retrieval monitoring to determine which control operations protect recall. The recognition analysis tests how presenting a candidate film item for judgment changes where reminder-linked task competition enters retrieval. Together, the simulations define a mechanistic alternative to selective-vulnerability accounts and identify conditions under which intrusive and voluntary remembering should appear more or less dissociated.

Figure 1

Modeled trauma-film paradigm and schematic selective-interference pattern. (A) The modeled trauma-film paradigm includes film encoding, an intervening delay, a possible pre-task film reminder, an intervening task, a post-task delay, and a later test of unguided or deliberate film recall. (B) The qualitative pattern targeted by the recall simulations is summarized as an interaction among reminder condition, intervening task, and retrieval condition. Columns distinguish whether the task is preceded by a film reminder, rows distinguish unguided film recall from deliberate film recall, and paired bars compare film recall after a comparison/control condition versus a visuospatial task condition. The visuospatial task condition produces the largest reduction in unguided film recall when it follows a film reminder, whereas deliberate film recall shows a smaller reduction under the same conditions. Bars are schematic and do not plot empirical counts, fitted values, or model output.

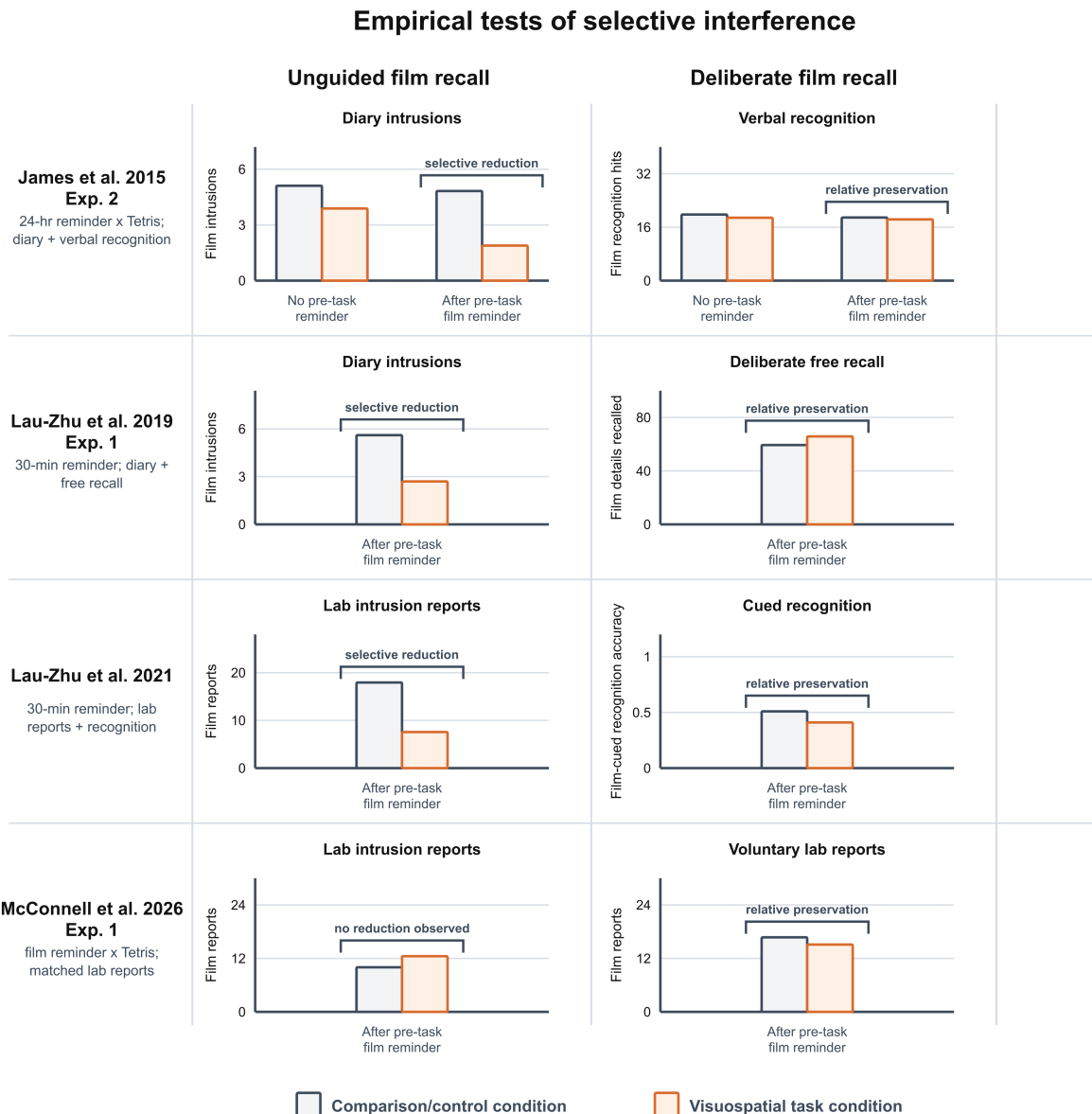


Intentionality and Test Format in Selective Interference

Selective-interference studies are often described as comparing involuntary and deliberate memory, but the empirical contrast also depends on how each form of remembering is measured. Diary measures count film memories after they inadvertently come to mind during daily life. Laboratory intrusion tasks count film memories reported during a defined monitoring period, sometimes while film cues are present. In both cases, participants report film memories when they occur without deliberately trying to recover film details. Voluntary-memory tests differ in whether target film material must be recovered or is presented for judgment: free recall and specific-probe questions ask participants to bring film details to mind, whereas recognition asks participants to distinguish true from false film statements or film stills from lure images (James et al., 2016; Lau-Zhu et al., 2019, 2021). Selective interference appears most consistent when intrusion reports are compared with recognition, and less uniform when film material must be brought to mind under more similar retrieval conditions. Figure 2 illustrates variation in retrieval format across selected trauma-film studies (James et al., 2015; Lau-Zhu et al., 2019, 2021; McConnell et al., 2026).

Figure 2

Empirical tests of selective interference across retrieval formats. Rows show selected studies that pair intrusion-like or involuntary-memory measures with different deliberate-memory tests: diary intrusions with recognition, diary intrusions with free recall, laboratory intrusion reports with recognition, and a matched voluntary/involuntary laboratory retrieval task (James et al., 2015; Lau-Zhu et al., 2019, 2021; McConnell et al., 2026). Left panels plot intrusion-like or involuntary-memory outcomes, and right panels plot deliberate-memory outcomes. Gray bars indicate the study-specific comparison/control condition and orange bars indicate the visuospatial task condition. Bracket labels summarize the plotted contrast; they do not denote a uniform significance test. Bar heights reproduce reported condition means in native study units, so y-axis scales differ across outcomes.



Comparisons between diary intrusion reports and recognition tests confound retrieval intentionality with retrieval format. In immediate post-film Tetris studies, participants completed Tetris soon after film viewing and later reported fewer diary intrusions, while verbal or visual recognition remained similar across groups (Holmes et al., 2009, 2010). Delayed reminder-plus-Tetris studies used a later film reminder before Tetris and found a related pattern when diary or laboratory intrusion measures were paired with recognition (James et al., 2015; Kessler et al., 2020). Later work pairing diary intrusions with recognition retained the same basic comparison (Asselbergs et al., 2024). In these studies, intrusion measures can count a film memory only after it comes to mind and is reported, whereas verbal and visual recognition tests present candidate film material for judgment (Lau-Zhu et al., 2019, 2021). A reduction in intrusions with preserved recognition therefore does not, by itself, show that voluntary memory is preserved across retrieval formats.

Evidence is less consistent in designs that make intrusion and voluntary-memory measures more comparable. Free recall and specific-probe questions make voluntary-memory performance depend on bringing film material to mind, matching a central requirement of intrusion reports (Lau-Zhu et al., 2019). Laboratory intrusion tasks collect reports during a defined period, making the setting more comparable to laboratory voluntary-memory tests than a week-long diary (Lau-Zhu et al., 2019, 2021). Across matched recall and laboratory-report formats, effects of Tetris after a film reminder are less uniform. Some designs still show reduced diary or laboratory intrusion reports with relatively preserved voluntary memory (Lau-Zhu et al., 2019, 2021). A stricter matched-report design did not find the predicted reduction in involuntary memory (McConnell et al., 2026).

Variation across retrieval formats motivates a mechanistic account of when and why selective interference appears. Recognition-based comparisons show selective interference more consistently. The same comparisons confound retrieval intentionality with test format. Attempts to compare recall-like measures across intentionality conditions are more methodologically robust, yet observe selective interference less consistently. A complete account must therefore

explain both how deliberate recall can remain preserved while unguided recall is reduced, as well as why selective interference depends on how retrieval is cued and measured.

A Retrieved-Context Account of Selective Interference

Figure 3

Encoding and associative structure in the retrieved-context account. (A) The model stores bidirectional item-context associations. Context-to-feature associations cue candidate items from the current context, and feature-to-context associations reinstate context when an item is recalled or when a reminder or test cue is presented. (B) The interference manipulation changes which context states task items are linked to during encoding. Without a pre-task reminder, task items are encoded after context has drifted away from the film. With a pre-task film reminder, film-associated context is reinstated before task encoding, so task items are learned while that context remains active. Thus, the same context that can cue film items can also cue task competitors during later retrieval. When task encoding is strong, those task items become stronger competitors during later context-guided retrieval. Circles mark item features: *F* labels film items and *T* labels task items. Squares mark context states. Blue marks film items or film-linked context, orange marks task items or task-linked context, green marks reminder cues, and gray marks neutral, delay, comparison, or shared model structure.

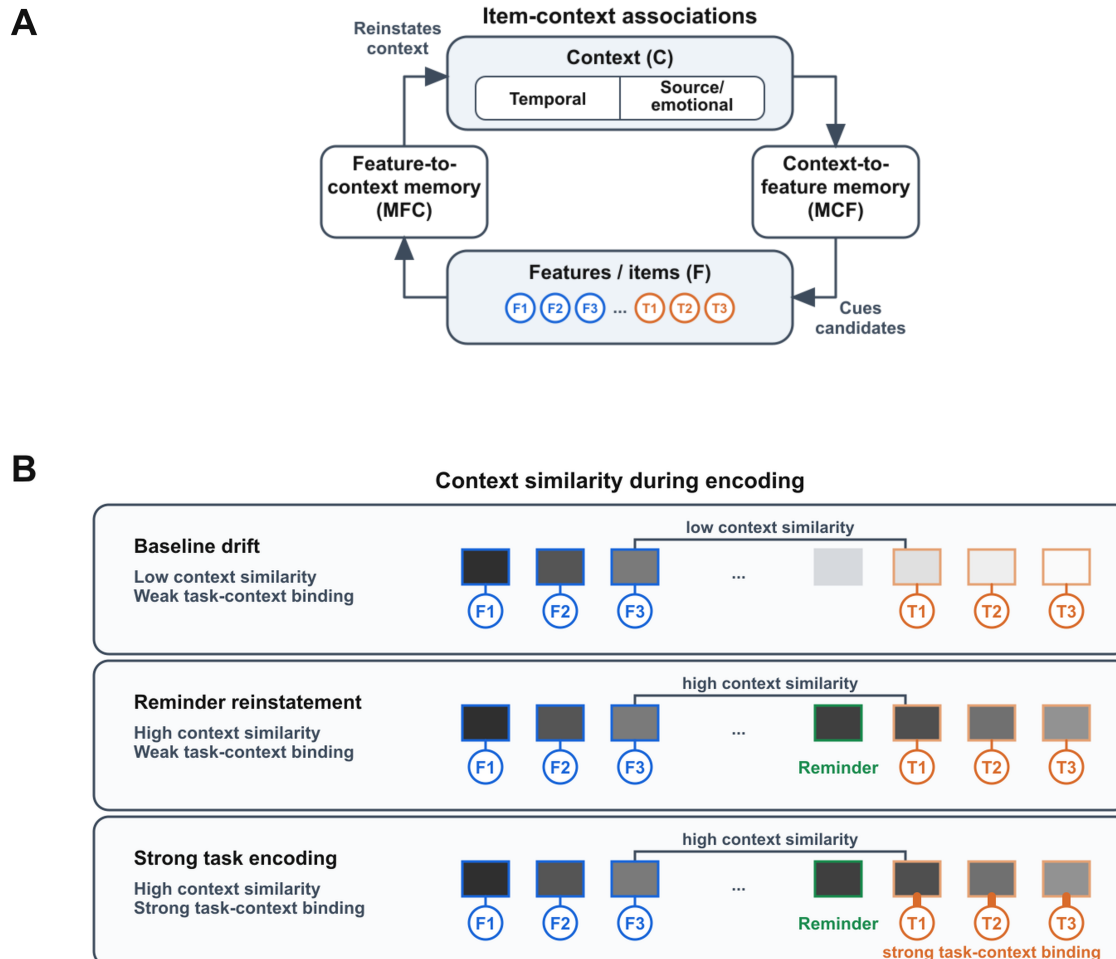
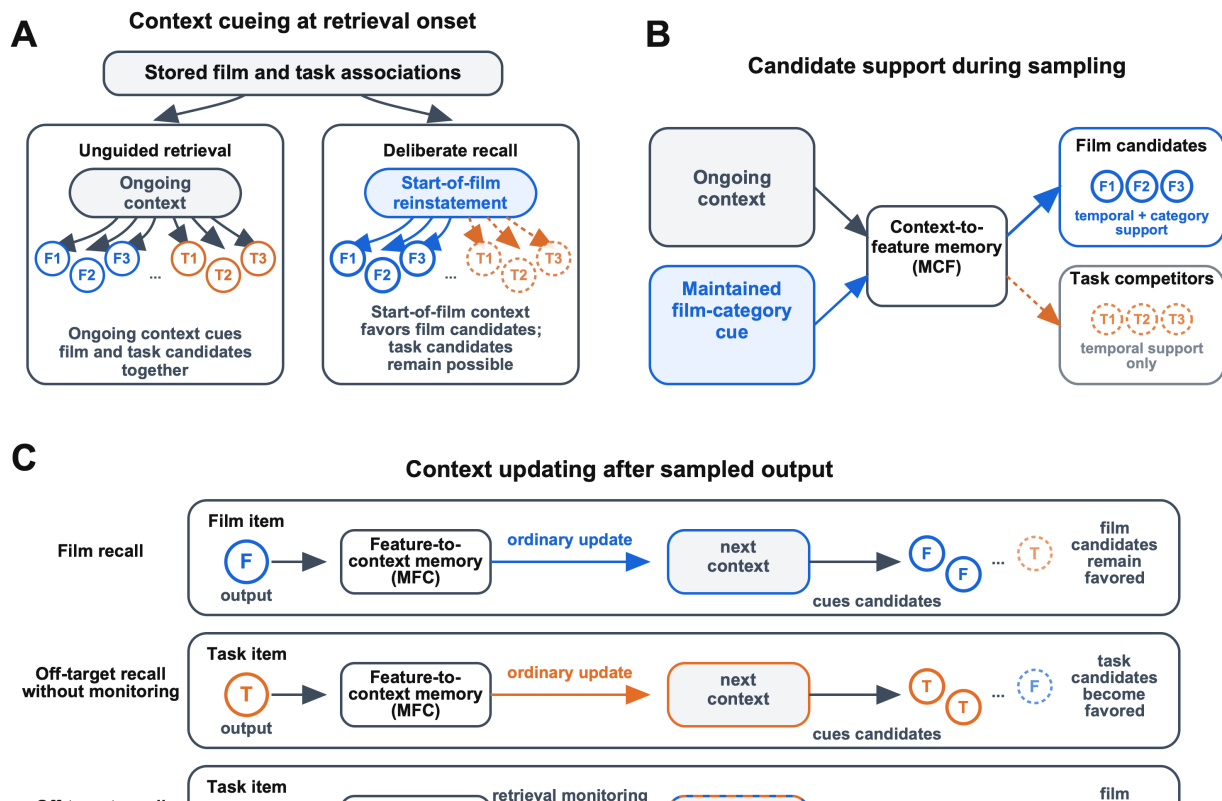


Figure 4

Retrieval operations in the retrieved-context account. (A) The upper box represents stored film and task associations; the lower boxes compare the context state used to begin unguided retrieval and deliberate recall. Stored film and task associations can be sampled from different starting contexts. Unguided retrieval begins from ongoing context, which can support film and task candidates together. Deliberate recall can begin with start-of-film context reinstatement, which favors film candidates while leaving task candidates possible. (B) During context-to-feature sampling, ongoing context can support film items and task competitors, while a maintained film-category cue adds support only to film candidates. (C) Rows compare film recall, unmonitored task recall, and monitored task recall. Sampled outputs update the next retrieval context through feature-to-context memory. Recalling a film item keeps later search biased toward film candidates. Recalling a task item without monitoring can shift later context toward task candidates. With retrieval monitoring, the task-item update is dampened, preserving some film-candidate support. Blue marks film items, film-biased context, or stronger film support; orange marks task items, task-biased context, or stronger task support; gray marks shared context and associative operations. Dashed outlines or arrows mark weaker support or dampened updating, not unavailable candidates.



Retrieved-context models explain recall as search from a changing context state shaped by recent experience and prior retrieval. The current context cues candidate items, and recalled items reinstate context from encoding, producing temporal organization in free recall (Howard & Kahana, 2002; Sederberg et al., 2008). Applied to selective interference, these operations explain how task material becomes linked to film-associated context and how later retrieval determines the extent to which that material competes with film material for recall.

Film, task, and delay periods are represented as sequences of events that update context. For the film, these events correspond roughly to scenes or segments. For the task and delay periods, they represent non-film events that can also be encoded and later sampled. A film reminder reinstates film-associated context before task encoding. Task events linked to that state can later be cued alongside film material. Retrieval then depends on how strongly the test setting constrains search toward film material rather than toward the broader set of candidates supported by the current context. Figure 3 summarizes the encoding and associative structure of the account, and Figure 4 summarizes the retrieval operations.

Context-Guided Search and Source Organization

Retrieved-context models were developed to explain structured regularities in open-ended recall. Free recall favors recently studied items, and successive recalls tend to come from nearby study positions (Howard & Kahana, 2002; Kahana, 1996; Sederberg et al., 2008). In these models, a gradually changing context state cues candidate items, and recalled items reinstate the contexts in which they were encoded. Recency follows because the retrieval context remains similar to contexts from recent study and recall events. Temporal contiguity follows because a recalled item reinstates context from its study position, which is similar to the contexts associated with nearby items.

Recall is also organized by shared content, task, and source. Semantically related items, items encoded under the same task, and items from the same source can cluster during recall (Lohnas et al., 2023; Morton & Polyn, 2016; Polyn et al., 2009a, 2009b, 2012). CMR captures semantic and source organization by allowing each form of information to contribute to the

context state that guides retrieval. Semantic and source dimensions can make a retrieval cue more selective than temporal context alone: a target-category or source cue can favor one subset of items even while temporal associations continue to organize retrieval within that subset (Lohnas et al., 2026; Polyn et al., 2011). Combining shared temporal support with selective non-temporal support is central to the retrieved-context explanation developed here.

Emotional memory gives source organization a further role in the model used here. Emotional items can cluster together, and emotional material is often remembered better than neutral material (Talmi et al., 2007; Talmi, 2013; Talmi & Moscovitch, 2004). Emotional CMR represents affective information as a source-like context dimension and allows emotional items to form stronger source-context associations (Cohen & Kahana, 2022; Talmi et al., 2019). Film events therefore share an affective/source-context signal while remaining distinct items with different temporal associations. Task and delay events remain available to the same sampling process and can acquire support from any temporal or source context active while they are encoded. The affective/source-context representation and the maintained film-category cue are separate model constructs. Neither construct formally implements the S-reps or C-reps in dual-representation theory.

Reminder-Linked Competitor Encoding

Reminder-linked competitor learning occurs when a film reminder reinstates context from the earlier event before task events are encoded. In the retrieved-context account, presented or retrieved items reinstate their associated context (Howard & Kahana, 2002; Polyn et al., 2009a; Sederberg et al., 2008). A film reminder therefore shifts the current state toward context from its corresponding film episode. If task events are then learned while reinstated film context remains active, they acquire context-to-item associations that overlap with associations supporting film events. Later film-oriented context can consequently support task candidates as well as film candidates.

Recall can include material outside the intended target. In multi-list free recall, participants sometimes report items from earlier lists instead of the list they were asked to recall

(Howard et al., 2008; Lohnas et al., 2015; Zaromb et al., 2006). Those errors are structured: they are biased toward recently studied prior lists and change as intervening study or delay changes. Retrieved-context models explain this pattern by treating recall as candidate sampling from the current context. As context drifts, support for non-target items declines. When a non-target item was encoded in a recent or related context, it can still enter the candidate set. This makes task and delay events eligible recall candidates in the trauma-film paradigm. If those events are associated with context states that later support film recall, they can compete with film material during retrieval.

Delayed intervention procedures in the trauma-film paradigm create the conditions for this kind of competitor encoding when a film reminder precedes the task. In these studies, participants encounter film cues before the interference task, and the procedure is typically interpreted as reactivating film memory before the task begins (James et al., 2015; Kessler et al., 2020). When participants return to an earlier learning context, new material can become linked to the earlier episode (Hupbach et al., 2008; Sederberg et al., 2011). The simulations model this sequence by placing task encoding after reminder-driven reinstatement of film-associated context. Task events encoded in that reinstated state acquire associations with context that also supports film items. The intervention can then reduce later film recall by adding task competitors while stored film-item associations remain available. Panel B Figure 3 summarizes this encoding claim: after a film reminder, task items acquire associations with context that can later cue both task competitors and film material.

Recognition and Presented Film Items

The retrieved-context framework distinguishes two routes by which film material can enter a test response. In context-guided retrieval, the current state of context cues candidate items, so a film memory is reported only if film material is selected from the items supported by that state. In item-guided retrieval, a presented film statement or still image is already the candidate, so the task is to evaluate the context retrieved by that item. Trauma-film studies commonly assess voluntary memory with recognition tests, where participants judge whether presented film statements or still

images appeared in the film (Lau-Zhu et al., 2019, 2021; Varma et al., 2024). By contrast, diary and laboratory monitoring tasks ask participants to report film memories that arise during daily life or during an ongoing task (Holmes & Bourne, 2008; James et al., 2016; Lau-Zhu et al., 2019). On this account, reduced intrusion reports with preserved recognition can reflect a test-format distinction rather than a dissociation between involuntary and voluntary remembering. Comparisons between intrusion-like reports and recognition tests confound retrieval intentionality with the route by which film material is accessed.

Recognition judgments are shaped by the context surrounding a test probe. Recently studied items are recognized more accurately than less recent items. Similar lures are harder to reject, and responses can reflect false familiarity, false recollection, or recollection rejection (Parks et al., 2026). One recognition probe can also influence recognition of items studied nearby (Healey & Kahana, 2016; Jin et al., 2024; Kahana, 2020; Schwartz et al., 2005). In retrieved-context accounts of recognition, a presented probe retrieves context associated with earlier encounters of that item, and the retrieved context is compared with the context active during the test (Healey & Kahana, 2016; Jin et al., 2024). Recency, lure similarity, and carryover between probes can be explained by changes in the match between the context retrieved by the probe and the context used for judgment. Recognition therefore probes memory with an item and evaluates the context that item retrieves.

Reminder-linked task encoding interferes with diary and monitoring measures when film-associated context cues non-film competitors. Those measures count film material only when it is retrieved during daily life or during a monitoring period. After a film reminder, task events can be encoded while film-associated context is active. Later, film-associated context can cue those task events along with film items, reducing the chance that film material is retrieved. Recognition uses the item-guided route described above. The tested film item can retrieve associated context without first being selected from film-associated context. Figure 13 formalizes this contrast between context-guided retrieval and recognition: task items linked to film-associated context can compete with film items when context cues candidate items, whereas

recognition begins with a candidate film item already presented for judgment.

Voluntary recall can be tested by means other than recognition. In vigilance-intrusion tasks, participants perform an ongoing task while film or foil cues appear, and they report film memories that arise during the task (Lau-Zhu et al., 2019, 2021). A voluntary version can keep the same film-cued task structure while instructing participants to search deliberately for film memories, while an involuntary version can instead more neutrally instruct participants to report any film memories that come to mind (McConnell et al., 2026). Both versions score film material retrieved during a recall period rather than recognition judgments about presented candidates.

Film cues in these tasks still shape retrieval. A film cue can reinstate context associated with the cued film material. If recall depends on maintaining film-associated context across the retrieval period, film cues can support recall in both monitoring and deliberate-search conditions. After reminder-linked task encoding, however, the same reinstated context may also support task material, enabling task items to compete with film material during context-guided recall. A recognition probe changes the access route more sharply because the probe is the candidate item to be judged and decisions are based on the context that item retrieves.

Recognition-intrusion comparisons therefore make confounds test format and retrieval intentionality in measures of the selective interference effect. Better-matched recall designs keep film material dependent on context-guided retrieval across intentionality conditions, with film cues modeled as inputs to the retrieval context. Under that design, differences in the effects of reminder-linked task encoding on film recall can be attributed to the way deliberate recall controls the course of memory search.

Film-Directed Recall Control Under Matched Retrieval Conditions

Competitor encoding makes film recall vulnerable when retrieval is guided by context that supports both film and task candidates. Diary and vigilance-intrusion measures count film memories when they come to mind, often during ongoing activity or monitoring rather than after a specific film item has been presented (Holmes & Bourne, 2008; Lau-Zhu et al., 2019, 2021). The most relevant voluntary comparison similarly requires film material to be called to mind

during a recall period rather than judged as a presented recognition candidate (Lau-Zhu et al., 2019; McConnell et al., 2026). Under this matched recall design, a film reminder can make task material accessible from the same retrieval state as film material by linking task events to reinstated film-associated context. Control processes can improve deliberate recall by orienting retrieval toward the film. They protect film recall from reminder-linked interference to the extent that they make retrieval less dependent on film-associated context that also supports task material.

Within this matched recall setting, Figure 4 separates three ways deliberate recall can differ from unguided retrieval after the same reminder-linked task encoding. Start-of-film context reinstatement acts before sampling by changing the initial context state. A maintained film-category cue acts during sampling by adding support to film candidates. Retrieval monitoring acts after sampling by reducing the carryover from off-target candidates. These control points make separable predictions because they change different parts of the retrieval cycle. The simulations test which control changes the effect of reminder-linked task encoding, rather than simply increasing film recall under all conditions.

Deliberate recall can begin from a context state biased by retrieval instructions. When people are asked to recall an earlier studied list after studying a more recent one, their early responses are often already biased toward the instructed list rather than showing a gradual search backward through recent-list items (Healey & Wahlheim, 2024; Lohnas et al., 2015). Recall also shows a reliable tendency to start near the beginning of a studied list, especially when the usual end-of-list advantage is weakened by a delay (Healey & Kahana, 2014; Herrema & Kahana, 2024; Howard & Kahana, 1999). Start-of-list context reinstatement mechanisms account for these patterns by evolving the current context toward the state associated with the beginning of the target list before retrieval begins (Healey & Wahlheim, 2024; Kragel et al., 2015; Lohnas et al., 2015; Morton & Polyn, 2016). In our retrieval control simulations, start-of-film reinstatement drifts the retrieval state toward the context present at the beginning of film encoding. This starting state favors early film candidates before ordinary context-guided sampling resumes, so its predicted benefit is tied to the temporal region that has been reinstated.

Retrieval monitoring acts after a candidate has been sampled. Evidence for this post-sampling control comes from externalized recall tasks where participants report every word that comes to mind and mark errors. These instructions reveal many more intrusions than standard recall, and many sampled intrusions are marked as errors, indicating that off-target candidates can enter retrieval without being accepted as target memories (Lohnas et al., 2015; Unsworth et al., 2010). Monitoring also carries forward across retrieval: when people fail to identify one response correctly, they are worse at rejecting the next intrusion (Unsworth et al., 2010). At the same time, sampled intrusions can pull search away from the target list, as intrusions tend to cluster with other intrusions from related prior-list or semantic contexts (Unsworth et al., 2010; Zaromb et al., 2006). Retrieved-context implementations have treated monitoring as a target-match filter while still allowing rejected items to update the next retrieval context (Lohnas et al., 2015). Our trauma-film simulations explore whether that post-sampling evaluation can also reduce carryover from sampled non-film candidates. A sampled non-film candidate still counts as a retrieval attempt, but its context update is dampened when it is evaluated as off-target. Monitoring can therefore help deliberate recall return to film material after non-film competition appears, while leaving the initial competition from reminder-linked task encoding in place.

A goal to retrieve the film can guide candidate sampling through category-level information. When people study mixed-category material and are asked to retrieve only one category, they can restrict recall to the target subset while temporal organization remains visible within that subset (Lohnas et al., 2026; Polyn et al., 2011). In this kind of task, category-selective activity tracks retrieval effort, remains active between reported recalls, and weakens before extra-category intrusions (Norman et al., 2017). Retrieved-context models provide a formal precedent for this kind of non-temporal support: semantic, task, and source information can contribute to candidate support alongside temporal context (Morton & Polyn, 2016; Polyn et al., 2009a, 2009b). Category-directed search has been formalized in associative memory models, and recent review work treats this as a natural but still underdeveloped direction for CMR (Lohnas et al., 2026; Raaijmakers & Shiffrin, 1980). The simulations use this logic in reduced form by

adding a maintained film-category cue during candidate sampling. This cue gives film candidates additional support without equally supporting task competitors. Task items linked to reinstated film-associated context can still compete through temporal context, but they do not receive this film-category support.

Control processes engaged during deliberate recall can therefore improve film recall overall without producing the relative preservation against reminder-linked task interference that defines the selective-interference effect. That relative preservation depends on whether deliberate recall can draw on support beyond film-associated temporal context after that context has also become supportive of task material. Start-of-film reinstatement and post-retrieval context monitoring improve overall film recall by reinstating or preserving film-associated context as a retrieval cue. Yet these mechanisms work by keeping film-associated context cues available, while reminder-linked task encoding interferes with film recall precisely by making such context cues less selective for film material. By contrast, a maintained film-category cue gives film candidates support from a cue that task competitors do not share. Under the tested conditions, this film-category support limits the impact of those task competitors enough to produce the selective-interference pattern. This specifies how deliberate recall can succeed when task encoding after a film reminder reduces unguided film recall: the maintained film-category cue helps distinguish film candidates from task competitors during search.

Model Specification

Retrieved-context models implement recall as a sequence of sampling events in which current context gives graded support to candidate items (Polyn et al., 2009a; Sederberg et al., 2008). To express selective interference as changes in what gets sampled, film and non-film events have to be candidates in the same sampling process. The simulations implement this requirement by representing film, task, and delay periods as sequences of events that update context. For the film, these events correspond roughly to scenes or segments. For the task periods, they represent task items that can become competitors when encoded after reminder-driven context reinstatement. For the delay periods, they represent filler items: non-film events that can also be

encoded and later sampled. Interference manifests as a shift in sampled outcomes: task or filler candidates occupy sampling events that could otherwise retrieve film material.

The formal model links this sampling process to separate item and context representations. Each encoded event at position i is represented by an item feature vector f_i , a one-hot vector that distinguishes that event from other encoded events in the simulated paradigm. Context c_i denotes the internal context state after event i has updated context. As film, task, and filler events occur, they update context and become associated with the context active around them. During recall, the current context state determines which candidate items receive support.

During encoding, items and states of context are linked through bidirectional associations represented in two matrices. Context-to-feature memory, M^{CF} , maps context states to item support, allowing current context to cue candidate items during recall. Feature-to-context memory, M^{FC} , maps item features to context states so that cued, reminded, or recalled items can reinstate associated context. All vectors are treated as column vectors, so $M^{FC} f_i$ is a context vector and $M^{CF} c_i$ is a feature-support vector.

Context and corresponding associations are further subdivided into temporal and source components. Temporal context distinguishes unique items so different film, task, and filler events can produce distinct context states even if items come from the same phase of the paradigm or share other features. Source context represents coarser information shared across related events, allowing recall to be shaped by task, semantic, or source relationships among items (Polyn et al., 2009a, 2009b). Emotional CMR variants distinguish emotional from neutral material in source context and give emotionally salient items stronger source-context associations (Cohen & Kahana, 2022; Talmi et al., 2019). In the present implementation, each item also has a source feature vector s_i . Film items share a film-source feature, whereas task and filler items carry a neutral/default source feature.

Encoding the Paradigm Sequence

Film, task, and filler events become later recall candidates through context updating and item-context learning. Each event updates context and becomes associated with the resulting

state. During later recall, candidate support depends on how strongly studied items are associated with the context states active during retrieval.

Before the simulated film starts, context is initialized to c_0 , a start state with no item-associated context units active. Memory matrices are initialized with pre-experimental associations that reflect baseline structure already present before the simulated paradigm begins. In M^{FC} , each item i can cue its paired context unit with weight $1 - \gamma$. In M^{CF} , each item is most strongly supported by its paired context unit, with weight δ . The other item-associated context units provide weaker background support to that item, with weight α . The start state c_0 initially provides no item support in M^{CF} . The initial entries of M^{FC} and M^{CF} are therefore:

$$M_{pre}^{FC}(j, i) = \begin{cases} 1 - \gamma & \text{if } j = i + 1 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

$$M_{pre}^{CF}(i, j) = \begin{cases} 0 & \text{if } j = 0 \\ \delta & \text{if } j = i + 1 \\ \alpha & \text{if } j > 0 \text{ and } j \neq i + 1 \end{cases} \quad (2)$$

When event i is encountered, its representation f_i retrieves context input through feature-to-context memory:

$$c_i^{\text{IN}} = M^{FC} f_i. \quad (3)$$

Context then drifts toward the normalized input. The coefficient ρ_i keeps context normalized after the update:

$$c_i = \rho_i c_{i-1} + \beta_{\text{enc}} \hat{c}_i^{\text{IN}}. \quad (4)$$

Each event also encodes associations between its item features and the current context state. Feature-to-context learning updates M^{FC} so later cues or sampled items can reinstate associated context:

$$\Delta M^{FC} = \gamma c_i f_i^\top. \quad (5)$$

Context-to-feature learning updates M^{CF} so later context can support the item during sampling:

$$\Delta M^{CF} = \lambda \phi_i f_i c_i^\top. \quad (6)$$

Here, γ controls feature-to-context learning while λ scales context-to-feature learning. To capture primacy effects in recall, the scale ϕ_i boosts context-to-feature learning for early items in the paradigm and decays across subsequent items.

Encoding events also have a source feature vector s_i that distinguishes film items from non-film items. The source component of context follows the same normalized-drift form as temporal context, with source drift rate β_{emot} . All items can form associations with the source context active for that event, scaled by a baseline source-learning term η_s . For film encoding events, the source-context portion of the context-to-feature update receives an additional ϕ_{emot} scale to reflect stronger encoding of emotional material in the trauma-film paradigm (Cohen & Kahana, 2022; Talmi et al., 2019). Task and filler events carry the neutral/default source feature, so they use the baseline source-learning scale without the added film-source term. After encoding, film, task, and filler candidates can all enter later recall through the temporal and source-context associations learned for each item.

Reminder-Linked Task Encoding

After the film-delay period, the simulated paradigm can present film reminders before the task begins. The reminder phase reinstates film-associated context before task items are encoded. Task items encoded from that state can later receive support for recall from context that also supports film material.

The reminder phase begins by updating context toward the start state of the film sequence. Let c_b denote the context state after the break period. The reminder-phase start update is:

$$c_{r,0} = \rho_{r,0}c_b + \beta_{\text{rem,start}}c_0. \quad (7)$$

The model then applies a sequence of film-item reminder cues. Each item from the film phase is presented as a reminder cue in the order it was encoded. Feature-to-context memory retrieves reminder input as in Equation 3, and context drifts toward that input using the normalized update in Equation 4 with the reminder drift rate. Reminder cues update context but do not become recallable items and do not add new context-to-feature learning. This isolates the reminder’s effect to reinstating film-associated context rather than adding new film-item associations.

Task events are then encoded by the ordinary event-learning rules with the reminder-updated context as the starting state. Task items carry the neutral/default source feature rather than the film-source feature, so they do not receive the added ϕ_{emot} term. To represent stronger or weaker task encoding, the simulations scale context-to-feature learning for task events by a task-encoding-strength factor m_{task} :

$$\Delta M_{\text{task}}^{CF} = m_{\text{task}}\lambda\phi_i f_i c_i^\top. \quad (8)$$

Task items become interfering competitors in later recall because they become associated with context states that can cue film material. Reminder-linked task encoding therefore changes how non-film items are cued by film-associated context, without weakening the film-item associations formed during film encoding.

Context-Guided Recall and Test-Phase Cues

In recall formats where film material must come to mind before it can be reported, the current context state assigns support to film, task, and filler candidates. At sampling step j , candidate support is computed by probing context-to-feature memory with the full context state:

$$a_{i,j} = [M^{CF}c_j]_i. \quad (9)$$

The context vector c_j includes both temporal and source-context components. Candidates already sampled in the current recall period are excluded from later sampling.

Sampling follows a Luce choice rule over the remaining candidates (Luce, 1959):

$$P(i \mid \text{continue}) = \frac{a_{i,j}^\tau}{\sum_{\text{remaining } k} a_{k,j}^\tau}. \quad (10)$$

Following likelihood-based CMR implementations, stopping is modeled as a position-dependent hazard independent of current candidate support (Kragel et al., 2015; Morton & Polyn, 2016):

$$P_{\text{stop}}(j) = \theta_s e^{j\theta_r}. \quad (11)$$

This value is bounded to remain a valid probability. Because stopping is independent of candidate support, interference is read from the identity of sampled items. A sampled task or filler item takes the place of a possible film sample.

After a candidate is sampled, feature-to-context memory updates context toward the sampled item’s associated context:

$$c_{j+1} = \rho_{\text{rec},j} c_j + \beta_{\text{rec}} \widehat{M^{FC} f_{r_j}}. \quad (12)$$

This update lets sampled material shape later search. A sampled film item can keep later sampling near film-associated context. A sampled task or filler item can pull later sampling toward non-film material.

Test-phase film cues use the same feature-to-context operation without becoming recall outputs. When a film cue q is presented, the cued item updates context through $M^{FC} f_q$. The cue is not counted as a recalled item, does not make the cued film item unavailable for later recall, and adds no new learning. Recall sampling then resumes from the updated context state.

Deliberate Recall Control

Deliberate recall uses the same sampling rule as unguided recall, with control settings applied before, during, or after sampling. The control settings change the retrieval cue, candidate support, or sampled-item carryover without changing the film and task associations learned during encoding. This lets the simulations test which retrieval operations increase film recall and which ones protect film recall from reminder-linked task competition.

Start-of-film context reinstatement drifts context toward the state present at the beginning of film encoding before the first sampling step. The retrieval context is drifted toward the context state used to begin the film sequence, with drift rate β_{start} . This can bias early sampling toward film candidates rather than starting from the ongoing context state produced by the final delay phase of the paradigm. This control mechanism can therefore increase film recall. However, because it still relies on film-associated context to cue candidates, it does not substantially protect deliberate recall from the interference effects of reminder-linked task encoding.

In retrieved-context models of externalized free recall, monitoring is implemented as a post-sampling match check: rejected candidates are withheld as target responses but still update the next retrieval context (Lohnas et al., 2015). The present simulations keep the post-sampling timing but change the consequence of sampled non-film candidates. Post-retrieval context monitoring reduces the context update after sampled task or filler candidates. Let $\kappa_i = 1$ for film candidates and $\kappa_i = \kappa$ for task and filler candidates. With monitoring, the sampled-item update becomes:

$$c_{j+1} = \rho_j(\kappa_i\beta_{\text{rec}})c_j + \kappa_i\beta_{\text{rec}}\widehat{M^{FC}f_{r_j}}. \quad (13)$$

When $\kappa < 1$, sampled non-film candidates still count as sampled outputs but context is updated less strongly toward their associated context. This reduces carryover from sampled task or filler candidates without changing the candidate support that made them available. Like start-of-film reinstatement, this operation does not by itself remove task support from film-associated context.

During deliberate recall, the model can maintain a film-category cue that is absent during unguided recall. This cue abstracts the kind of semantic or category support used in retrieved-context models of selective recall, rather than updating temporal or source context during retrieval (Morton & Polyn, 2016; Polyn et al., 2011). The cue adds support to candidates marked as film items during sampling:

$$a_{i,j}^{\text{cat}} = a_{i,j} + \omega_{\text{cat}} g_i. \quad (14)$$

Here, $g_i = 1$ for film candidates and $g_i = 0$ for task and filler candidates. The indicator g_i is a retrieval-control cue, distinct from the source feature vector s_i used during encoding. Task and filler candidates can still compete through $a_{i,j}$, but only film candidates receive the film-category increment. Therefore, unlike start-of-film reinstatement and retrieval monitoring, a maintained film-category cue can reduce the impact of reminder-linked task encoding on film recall.

The three evaluated control mechanisms therefore enter the same retrieval cycle at different points: start-of-film context reinstatement before sampling, a maintained film-category cue during sampling, and retrieval monitoring after sampling.

Item Recognition

Recognition begins with a presented candidate item rather than a sampled recall candidate. A presented film item retrieves associated context through the same feature-to-context associations used by reminders, cues, and sampled items. Following retrieved-context accounts of recognition, recognition evidence is the match between the context retrieved by the probe and the context active during the test (Healey & Kahana, 2016; Jin et al., 2024; Lohnas et al., 2015):

$$e_q = (M^{FC} f_q)^\top c_{\text{test}}. \quad (15)$$

This recognition implementation isolates the access difference between recognition and context-guided recall. Recall requires film material to be selected from context-supported candidates, whereas recognition begins with a presented candidate item and evaluates the context it retrieves. The simulation stops at recognition evidence because that evidence is sufficient for the

present claim: a presented film item can retrieve associated context without first competing with task items for selection. The model therefore addresses whether task competitors linked to reinstated film-associated context disrupt item-guided access to film context, not how recognition evidence is converted into decisions or how other interventions might disrupt performance.

Simulations

The simulations examine a schematic delayed reminder-plus-task trauma-film paradigm, with James et al. (2015) and Kessler et al. (2020) providing procedural examples rather than fitted targets. The reminder-before-task sequence places task events after reminder-driven reinstatement of film-associated context, making it possible to examine whether those events become competitors during later retrieval. The core recall target is the model analogue of selective interference: strong task encoding following a film reminder reduces unguided film recall more than deliberate film recall. Unguided recall provides a simplified model of film material coming to mind without deliberate intent to retrieve it; it does not reproduce diary or laboratory intrusion procedures. Simulation 1 shows how a task performed after a film reminder attenuates film recall when task items become strongly linked to reinstated film-associated context. Simulation 2 adds the intentionality manipulation by contrasting unguided retrieval with deliberate recall and examining control mechanisms that could protect film recall. The recognition simulation then examines whether interference is reduced when the test presents a candidate film item for judgment rather than requiring that item to be sampled from context. Together, these simulations show how selective interference can arise from competitor encoding after reminder-driven context reinstatement and retrieval conditions that differ in how they access or support film material.

Methodology

The simulations require default values for context drift, associative learning, sampling, and stopping before the trauma-film manipulations are applied. Estimating those values from selective-interference data would require the full modeled paradigm together with trial-by-trial recall sequences that identify which film, task, and non-film candidates entered retrieval. Published trauma-film studies motivate this qualitative target, but they typically report intrusion

counts or final memory scores rather than the sequence-level data needed to estimate retrieval competition. We therefore use an immediate affective free-recall dataset to set the model's core encoding, retrieval, and stopping parameters.

In that dataset, participants who reported concern about the war in Ukraine studied and recalled lists of war-related and neutral photographs ([Dupertuys et al., 2026](#)). The fitted dataset included 123 participants, four lists per participant, 492 immediate recall sequences, and 3,804 scored recall events. Parameters were estimated from the ordered recall sequences using the model likelihood, yielding a pooled parameter regime rather than separate fits to the selective-interference simulations. The resulting parameter values provide a starting regime for context-guided recall of affective and neutral photographs.

The selective-interference simulations then add the trauma-film phase structure and manipulations of reminder presence, task encoding strength, and retrieval control. Additional implementation settings adapt this regime to the trauma-film sequence. Film items reinstate the marked source feature into source context, and neutral/default events can weakly bind active source context. The primacy profile is also moderated relative to the immediate-recall fit so that the shorter film sequence is not dominated by an unrealistically steep first-item advantage. [Table 2](#) gives the default numerical values used in the simulations.

In all simulations, the model first encodes the paradigm events and then enters retrieval. Film, break/buffer, task, and filler items are encoded by the CMR operations defined above. Each simulated trial contains 16 film items, 16 break/buffer items, 16 task items, and 16 filler items unless otherwise noted. Break/buffer items implement the delay before the reminder or task, and filler items implement the post-task delay. Equal phase lengths make redistribution across film, task, and filler periods directly comparable.

Weak film cues are included during recall to approximate laboratory retrieval settings in which film stills or blurred film images appear while participants monitor for, or deliberately search for, film memories ([Lau-Zhu et al., 2019, 2021](#); [McConnell et al., 2026](#)). They are held constant across recall conditions so that condition differences reflect the pre-task reminder,

task-encoding strength, and retrieval-control manipulations, rather than differences in test-phase cueing. Reminder cues and test-phase film cues update context through Equation 3 and Equation 4, but they do not add recallable items or new item-context learning. Simulations that present candidate item information in a different test format specify that route separately. In context-to-item recall simulations, all encoded film, break/buffer, task, and filler items can enter the retrieval competition. Retrieval allows up to 48 sampling steps, and sampled items are removed from the candidate set so the same item is not sampled twice. The simulated retrieval sequence records sampled candidates by type and encoded position, rather than only final film counts. This convention makes it possible to examine predictions about how interference redistributes recall across the encoded stream, rather than only reporting final recall counts.

Parameter sweeps rerun the full encode-then-retrieve simulation across planned values of a manipulated parameter. For each condition and manipulated parameter value, the model first completes the phase sequence, then enters retrieval and samples recallable items using Equation 10 and Equation 11. This procedure is repeated across simulated trials within each cell. When figures plot recall across the encoded stream, they summarize retrieval after the full phase sequence has been encoded. Serial-position curves plot the proportion of trials in which each encoding position was retrieved, and phase summaries average those proportions within each phase. Thus, position and phase labels identify encoding positions, whereas recall probabilities refer to subsequent retrieval.

Simulation 1: Task Encoding and Context Similarity

Simulation 1 tests whether reminder-linked task encoding can create competitors for unguided film recall. Retrieval settings are held fixed, so differences across conditions reflect associations formed before retrieval rather than deliberate control. The design crosses pre-task reminder presence with task encoding strength while weak film cues are supplied periodically during recall in all conditions. In the no-reminder condition, the reminder-start and film-cue drift scales are set to zero. In the with-reminder condition, context first shifts toward the start state and then receives sequential film reminder cues before task encoding. Reminder cues update context

without applying Equation 5 or Equation 6 and without adding recallable items. Task encoding strength is modeled as a multiplier on task-phase context-to-feature learning in Equation 6.

Figure 5**Simulation 1: unguided film recall by reminder condition and task encoding strength.**

Reminder condition and task encoding strength were crossed while retrieval mode was held unguided; weak film cues were supplied periodically during recall in every condition. **(A)** Example generated recall sequences for each condition. Each row shows one simulated trial selected near the condition mean for the number of film items recalled; recalled items are ordered left to right by output position. Colors mark film items, task items, delay/filler items, and unfilled output positions. **(B)** Mean number of film items recalled across all simulated trials. The bracket marks the larger reduction from weak to strong task encoding when the task follows a film reminder. Film recall was lowest when a film reminder preceded strong task encoding, isolating the unguided-retrieval component of the selective interference effect.

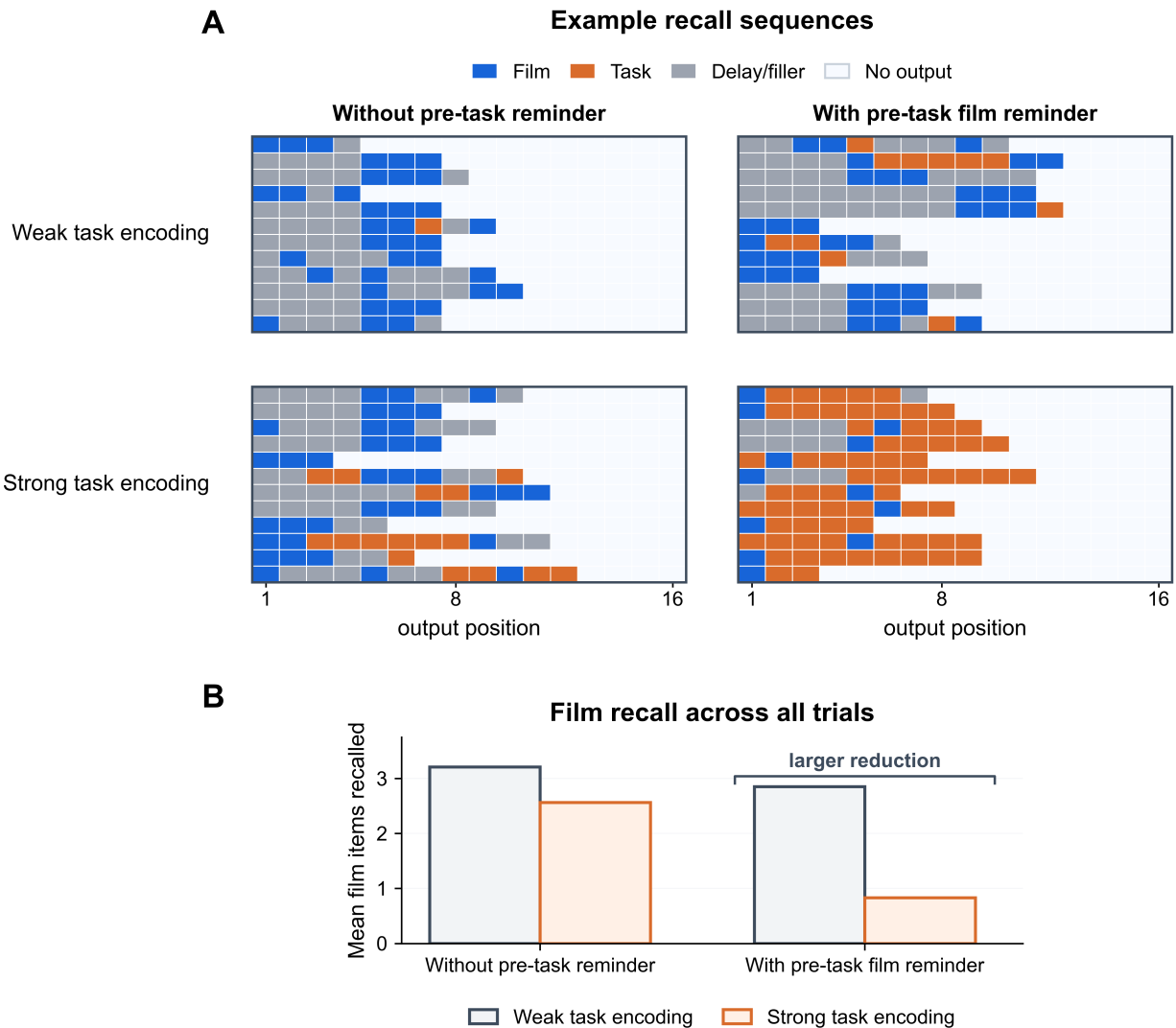


Figure 5 shows the aggregate result. Film recall is lowest when strong task encoding follows a film reminder, and task output increases in the same condition. The film loss therefore reflects redistribution toward task competitors under fixed retrieval settings. Figure 6 shows that the redistribution is position-resolved. Without a pre-task film reminder, increasing the task encoding multiplier produces only modest film loss and task gain, because task items become associated with later context states reached after drift away from the film. With a pre-task film reminder, the same task-encoding manipulation produces a larger drop in film-position recall and a larger rise in task-position recall. In the selected weak-versus-strong comparison, mean film recall in the no-reminder condition changes from 3.211 to 2.562 items, while mean task recall changes from 0.143 to 2.105 items. In the condition with a pre-task film reminder, mean film recall changes from 2.852 to 0.830 items, while mean task recall changes from 0.987 to 5.282 items.

Figure 6

Simulation 1: reminder-linked task encoding redistributes recall probability by encoded position. Reminder condition and task encoding strength were crossed while retrieval was held unguided; weak film cues were supplied periodically during recall in all conditions. **(A)** Without a pre-task film reminder, recall probability by encoded position changes only modestly when task encoding is strengthened. **(B)** With a pre-task film reminder, strong task encoding reduces recall from film positions and increases recall from task positions. Curves compare weak and strong task encoding; shaded bands mark film, break, task, and filler positions, and the green vertical line marks the pre-task film reminder. Task encoding strength therefore changes recall probability by encoded position most clearly when task encoding follows a film reminder.

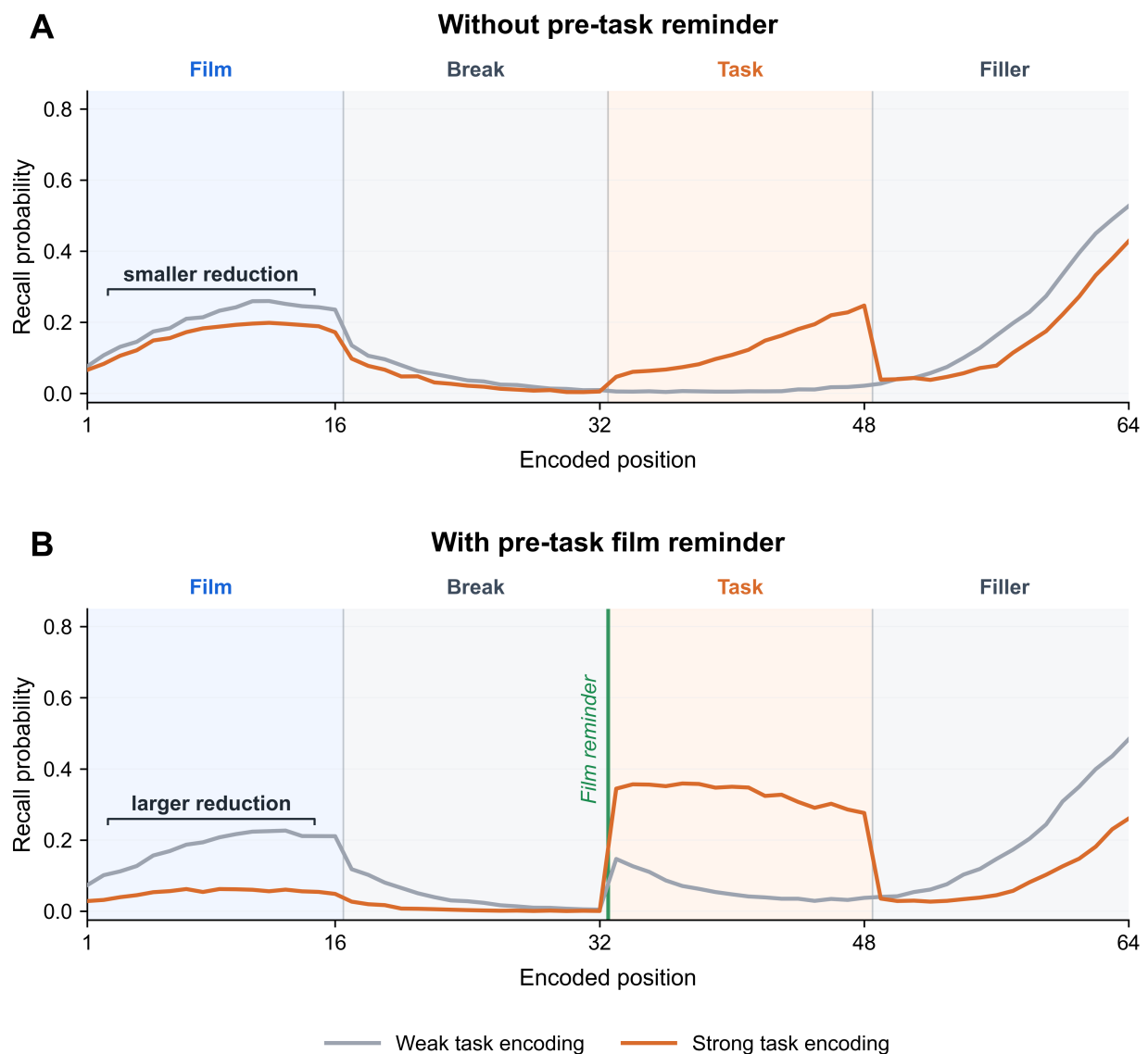


Figure 7

Simulation 1: reminder reinstatement increases similarity between film- and task-encoding contexts. Reminder condition was varied before task encoding while retrieval was held unguided.

(A) Mean context similarity between each encoded film and task item position, averaging across encoded task positions. (B) Full film-by-task context-similarity matrices for the same conditions; each cell compares the context state associated with one encoded film position to the context state active at one encoded task position. Brighter cells indicate greater context similarity. The film reminder increases similarity between film-encoding and task-encoding contexts, especially for later film positions and early task positions.

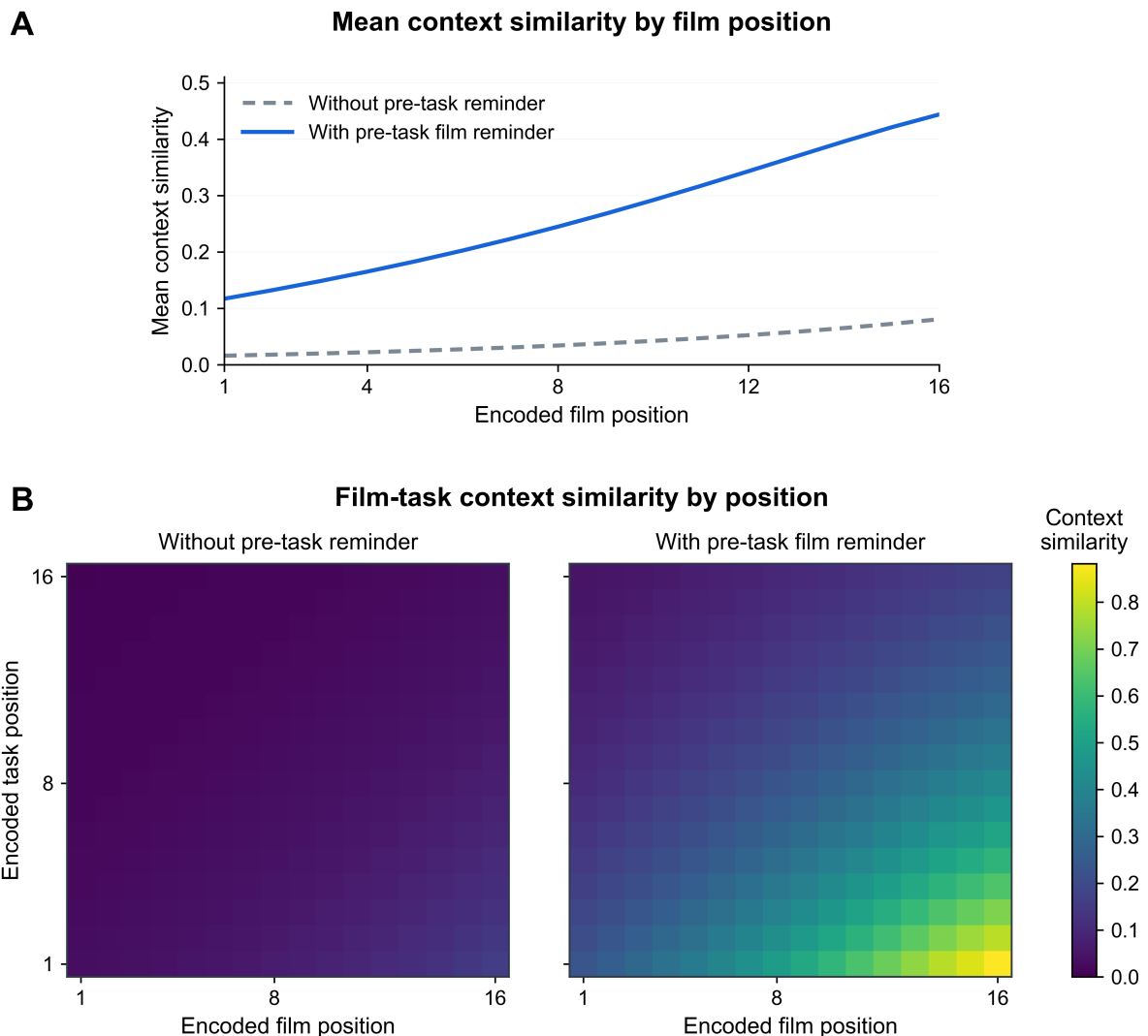


Figure 7 helps interpret why the no-reminder and with-reminder curves diverge. Sequential film reminders do not reinstate a single context state for the whole film. They reinstate film-associated states before task encoding, increasing similarity between task-encoding contexts and the context states active at particular film positions. Task encoding strength then determines how strongly task items are learned in those reinstated states, while film encoding settings remain unchanged. Task items remain distinct from film items, and film-item associations are not weakened. The loss of film recall occurs because task items become more likely to be sampled from context states that also support film material. Simulation 1 therefore establishes a retrieved-context account of how reminder-linked task encoding can create competitors for unguided film recall. Simulation 2 examines whether deliberate retrieval control reduces the effect of those learned competitors.

Simulation 2: Control Mechanisms for Deliberate Recall

Simulation 2 examines how deliberate retrieval control can protect film recall across the same reminder conditions and task encoding strengths used in Simulation 1. It first compares unguided and deliberate film recall across that design, then decomposes the control operations available during deliberate recall. The full deliberate-control condition combines three operations: start-of-film context reinstatement before sampling, a maintained film-category cue during sampling, and retrieval monitoring after non-film candidates are sampled. The main readout is how much strengthening task encoding reduces film recall in each reminder condition, rather than total film recall alone.

Figure 8***Simulation 2: deliberate film recall is protected from reminder-linked task interference.***

Reminder condition and task encoding strength were crossed as in Simulation 1 while retrieval mode was varied between unguided film recall and deliberate film recall. Deliberate film recall combined start-of-film context reinstatement, a maintained film-category cue, and retrieval monitoring. Weak film cues were supplied periodically during recall in all conditions. Rows compare retrieval modes, columns compare reminder conditions, and paired bars compare weak versus strong task encoding. In the with-reminder column, brackets label the selective reduction under unguided film recall and relative preservation under deliberate film recall. Strong task encoding reduced film recall most clearly when it followed a film reminder, but this reduction was much larger under unguided film recall than under deliberate film recall. The model therefore simulates a selective interference effect: strong task encoding after a film reminder substantially reduces unguided film recall while leaving deliberate film recall relatively preserved.

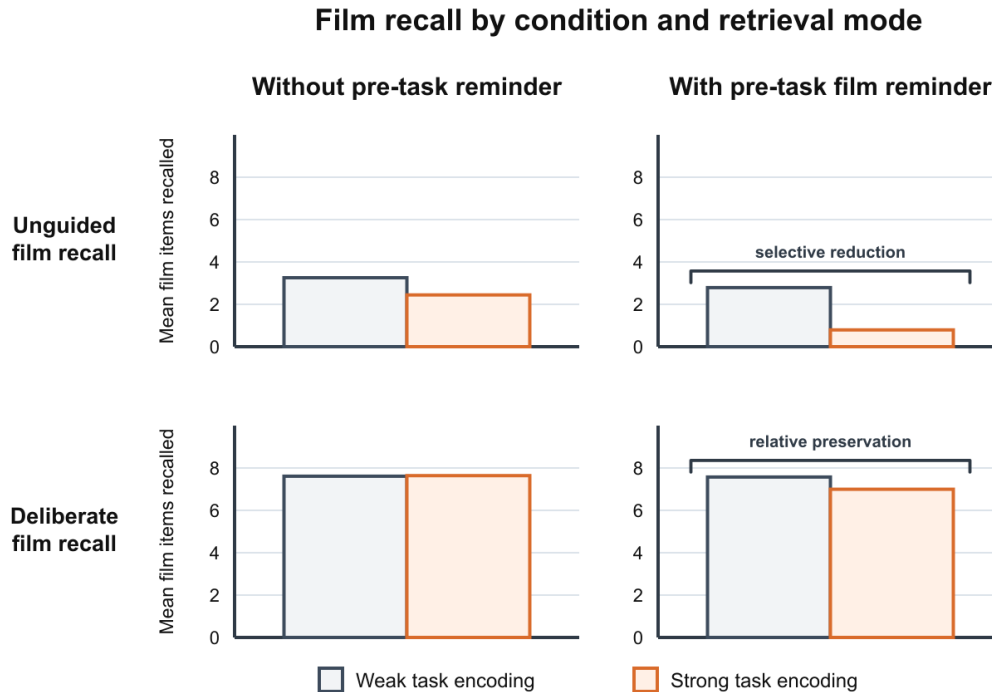


Figure 8 demonstrates the model's simulation of the selective interference effect. In

unguided film recall, strong task encoding produces a small film-recall loss without a pre-task reminder and a much larger loss after a film reminder. Under deliberate film recall, strong task encoding after a film reminder produces a much smaller reduction. Mean film recall in the with-reminder condition drops from 2.790 to 0.793 items under unguided film recall, but from 7.572 to 6.995 items under deliberate film recall. The full deliberate-control setting therefore reproduces the selective-interference pattern under the same learned task competitors.

Figure 9

Simulation 2: different retrieval controls preserve film recall to different degrees. Each control setting was evaluated in two conditions: a low-interference condition with no pre-task reminder and weak task encoding, and a high-interference condition with a pre-task film reminder and strong task encoding. Weak film cues were supplied periodically during recall in all conditions. Unguided film recall includes none of the three deliberate-control operations. Bars show mean number of film items recalled for each control setting in both conditions. Start-of-film reinstatement and retrieval monitoring increase film recall in some conditions, but the loss from the low- to high-interference condition is smallest when a maintained film-category cue is included.

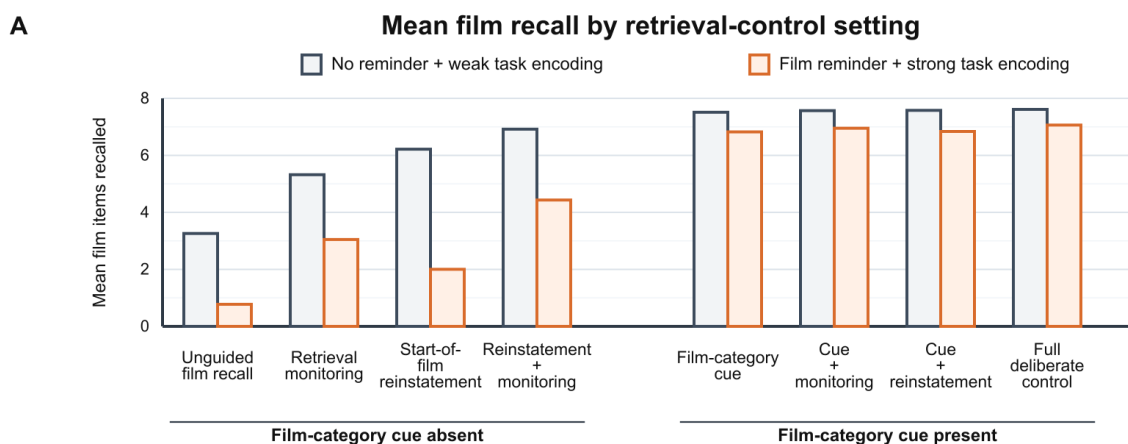


Figure 9 decomposes how retrieval-control operations alter the film-recall loss produced

when a film reminder precedes strong task encoding. The figure compares film recall in a low-interference condition, with no reminder and weak task encoding, against film recall in a high-interference condition, where a film reminder precedes strong task encoding. When the maintained film-category cue is absent, film recall remains much lower in the high-interference condition. Start-of-film reinstatement and retrieval monitoring can raise film recall, but the loss between the low- and high-interference conditions remains large. When a maintained film-category cue is active, the loss remains below one film item across the tested control packages; with all three controls present, film recall drops by about half an item. The maintained film-category cue most strongly reduces this loss by helping film candidates overcome competition from task competitors.

Figure 10

Simulation 2: a maintained film-category cue makes film-directed retrieval cues less similar to task-encoding context. The high-interference condition was held fixed: a film reminder preceded strong task encoding, and weak film cues were supplied periodically during recall. A maintained film-category cue supports film items during deliberate recall and is distinct from the external film reminder and test-phase film cues. **(A)** Mean similarity between task-encoding contexts and film-directed retrieval cues, averaging across encoded task positions for each encoded film position. **(B)** Full similarity matrices for the same condition. The left matrix uses temporal context alone; the right matrix schematically adds the maintained film-category cue to the film-directed retrieval cue. Brighter cells indicate greater similarity to task-encoding context. Adding the maintained film-category cue reduces similarity between film-directed retrieval cues and task-encoding contexts, providing a route to film recall that depends less on temporal context remaining selective for film items.

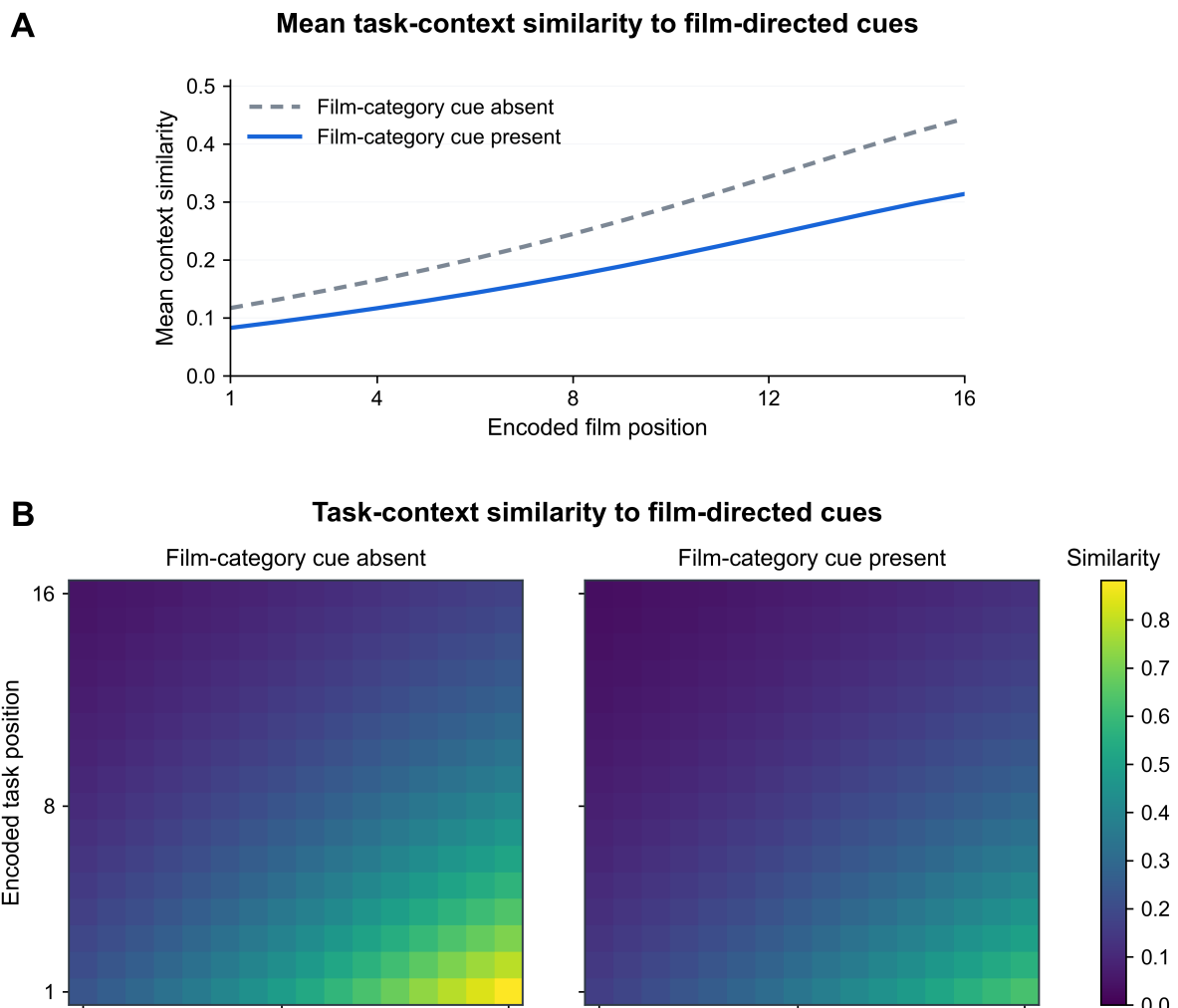


Figure 11***Simulation 2: start-of-film context reinstatement protects early encoded film positions.***

Unguided recall was compared with start-of-film context reinstatement while the condition with a pre-task film reminder and strong task encoding was held fixed and retrieval monitoring was disabled. Start-of-film context reinstatement initializes retrieval by drifting the current context toward the state present at the start of film encoding. Weak film cues were supplied periodically during recall. Shaded bands mark film, break, task, and filler positions; the green line marks the reminder immediately before the task phase. (A) Item accessibility before the first recall attempt. Reinstatement increases accessibility most strongly for early encoded film positions. (B) First recall probability distribution. Reinstatement shifts recall initiation toward early film positions. (C) Full recall probability distribution. The early-film bias remains visible across the recall period, while later film positions remain more vulnerable when strong task encoding follows a film reminder. Start-of-film context reinstatement therefore protects early encoded film positions by shifting retrieval initiation toward the start of the film sequence.

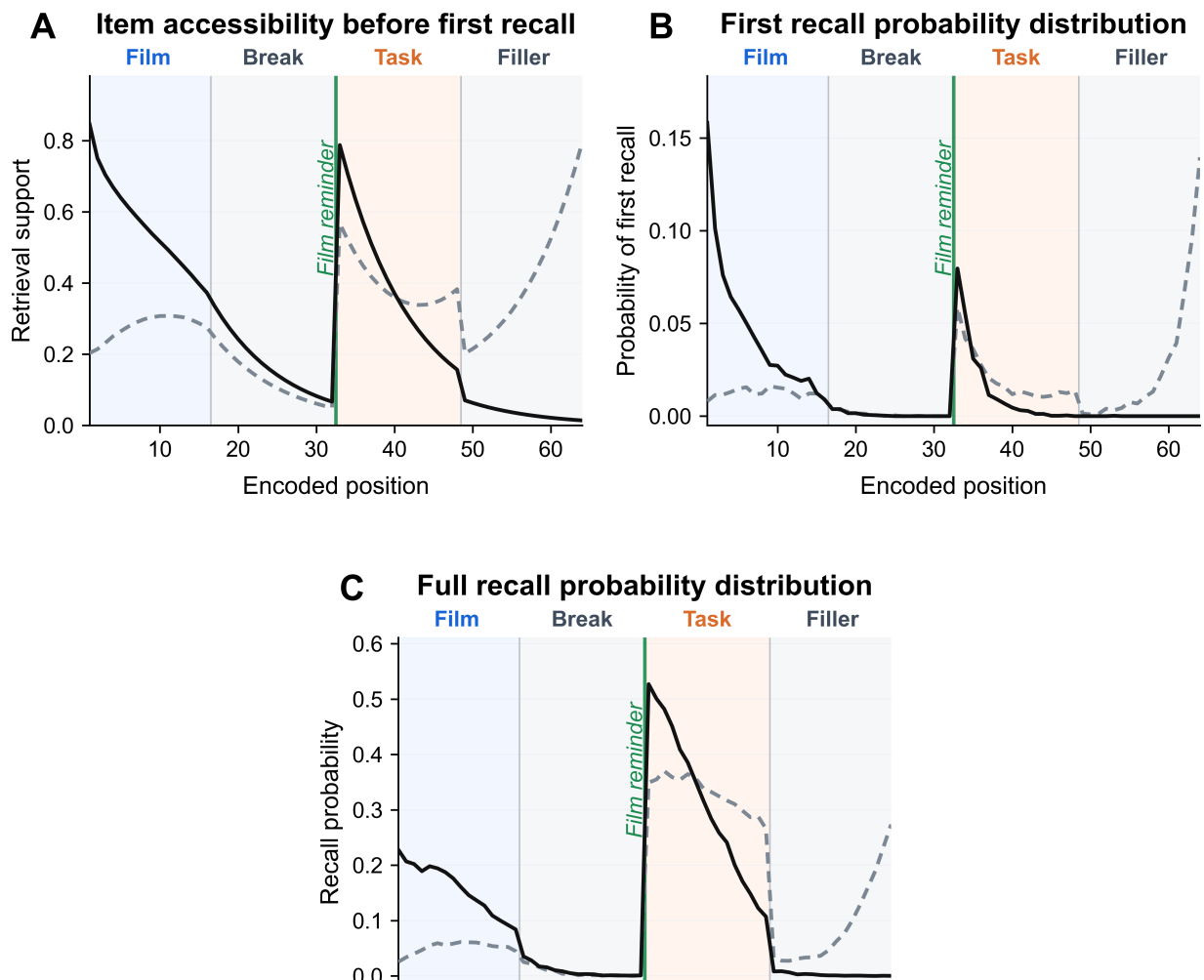
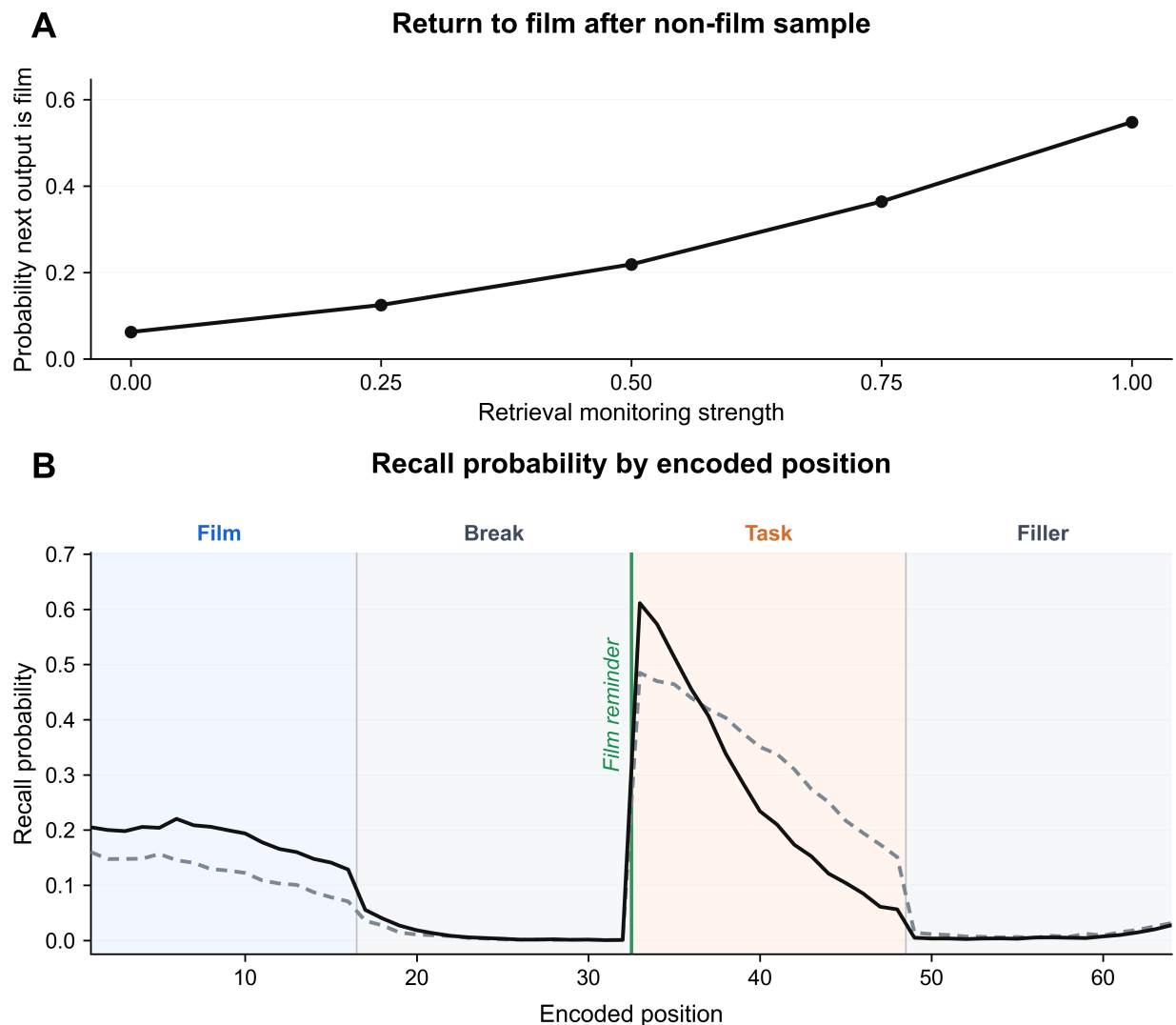


Figure 11 holds the high-interference condition fixed and isolates start-of-film context reinstatement. Reinstatement increases access to early film positions and shifts the first sampled item toward the beginning of the film sequence. Those gains persist in the full recall curve, but later film positions remain vulnerable because the start state does not remove task support from the contexts that cue later film items. The model therefore predicts that start-of-film context reinstatement will preferentially protect early film material, while later film material remains vulnerable to reminder-linked task competition.

Retrieval monitoring acts after a non-film item has already been sampled. By dampening that item's context update, monitoring limits how much the sample redirects subsequent retrieval. Figure 12 shows that monitoring increases the probability that a sampled non-film item is followed by recall of a film item. Monitoring therefore helps recall recover after task or filler material has entered the output sequence. It does not supply film-directed support before sampling occurs, so it provides only limited protection against task competition during film recall. Simulation 2 therefore constrains the account: deliberate recall is not spared by a generic voluntary-memory setting. It is less affected when control supplies film-directed support that task competitors do not share, whereas controls that only change retrieval initiation or post-sampling carryover remain more vulnerable.

Figure 12

Simulation 2: retrieval monitoring supports return to film recall. The condition with a pre-task film reminder and strong task encoding was held fixed, start-of-film context reinstatement was present, and retrieval monitoring strength was varied. Retrieval monitoring dampens context updating after non-film samples during deliberate recall. Weak film cues were supplied periodically during recall. **(A)** Model-diagnostic probability that a non-film sample was followed by recall of a film item as retrieval monitoring increased. **(B)** Recall probability by encoded position with no retrieval monitoring versus the moderate retrieval-monitoring setting used in the Simulation 2 anchor figure. Shaded bands mark film, break, task, and filler positions; the green line marks the reminder immediately before the task phase. Retrieval monitoring therefore supports film recall after search is pulled away from the film sequence, increasing film recall probability when strong task encoding follows a film reminder.



Simulation 3: Recognition

The recognition simulation uses the same reminder conditions, task encoding strengths, and resulting item-context associations as the recall simulations. It changes the retrieval operation by presenting each candidate film item as a test probe rather than requiring that item to be sampled from context. Context-guided recall requires a film item to be sampled from the candidates supported by current context. Recognition begins with a presented candidate item and asks whether the context retrieved by that item supports its occurrence in the film.

The simulation presents studied film items and unstudied foil identities that carry the same film-source feature as the studied items. Because item features are localist, these film-source-matched foils do not represent graded perceptual or semantic similarity to studied film items. The source-matched foils provide a baseline with the same broad film-source feature but no study-specific item-context association. This permits the narrower test of whether linking task items to film-associated context reduces the evidence advantage for studied film probes. Each probe retrieves associated temporal and source context through feature-to-context associations, and recognition evidence is computed from the match between the probe-retrieved context and the current test context (Healey & Kahana, 2016; Jin et al., 2024). The simulation estimates recognition evidence after the candidate item has already been presented as a probe. Decision criteria, confidence, and detailed foil-processing assumptions are left outside the present implementation.

Figure 13

Item-to-context retrieval bypasses reminder-linked task interference. (A) CMR stores bidirectional associations between item features and context. Feature-to-context memory lets an item reinstate its associated context, whereas context-to-feature memory lets context cue candidate items. (B) When film-associated context cues items, film items and task items linked to that context can both receive support. When a film item reinstates context, retrieval targets context units instead, so task items are not competitors in that step. (C) In recognition, the test presents a candidate film item as the probe. The probe retrieves associated context, which is compared with ongoing test context; stronger matches provide stronger evidence that the item appeared in the film. (D) Old-film minus foil recognition evidence is plotted across the same reminder conditions and task encoding strengths used in the recall simulations. The measure subtracts mean evidence for film-source-matched foils from mean evidence for studied film probes. Strong task encoding after a film reminder does not produce a corresponding reduction in recognition-evidence separation.

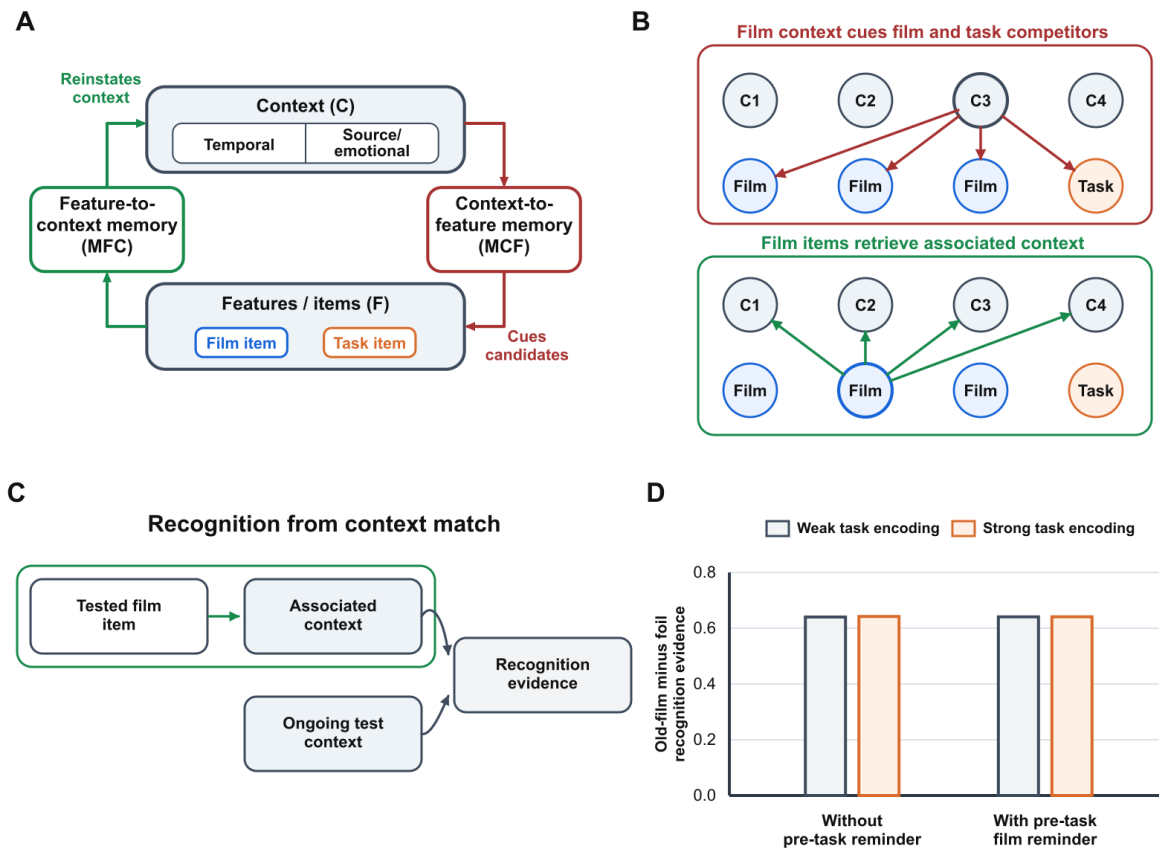


Figure 13 shows the resulting test-format consequence. Strong task encoding after a film reminder strongly reduces context-guided film recall but does not produce a corresponding reduction in recognition-evidence separation. Old-film minus foil evidence remains essentially unchanged across the four combinations of reminder presence and task encoding strength, ranging from .640 to .642. In context-guided recall, film and task candidates can compete whenever the retrieval context supports both. Recognition reverses that access route: the presented film candidate retrieves associated context directly, so task items do not compete to be retrieved in place of the film item. The simulation therefore identifies a test-format confound in designs that compare involuntary tests measuring whether film material comes to mind with voluntary tests that supply film material for recognition. Such designs therefore do not isolate retrieval intentionality: preserved recognition alongside reduced intrusions may reflect a difference between test formats rather than a general sparing of voluntary recall.

General Discussion

The simulations identify a condition under which goal-directed remembering can remain selective after target-associated context has become linked to competitors. A film reminder made subsequently encoded task items retrievable from film-associated context, reducing film recall when search relied on that context. Start-of-film reinstatement and retrieval monitoring improved film recall, but neither made recall comparably resistant to the learned task competitors. The largest relative protection occurred when deliberate recall included a maintained film-category cue that supported film candidates without equally supporting task competitors. Recognition produced a separate form of preservation because the candidate film item was presented before its associated context was evaluated. Together, these results locate selective interference in the relation between what has been bound to context and how later retrieval selects or evaluates film material.

From Contextual Competition to Selective Retrieval

Context binding creates both access and interference. Reinstating a prior context before new learning can link later material to the earlier episode, allowing the earlier context to cue both sets of events (Hupbach et al., 2008; Sederberg et al., 2011). Film-associated context can therefore

remain available while becoming less diagnostic of film material. Research on category-directed recall provides a complementary principle: people can maintain semantic or category information that favors target memories while temporal associations continue to organize retrieval (Lohnas et al., 2026; Norman et al., 2017; Polyn et al., 2011). Related retrieval-orientation findings indicate that this selectivity depends jointly on internal goals and whether available cues discriminate targets from competing traces (Moccia et al., 2025; Moccia & Morcom, 2022). Combined, these findings suggest that goal-directed remembering remains selective when retrieval can supplement a shared contextual cue with target-defining information that competitors do not share.

This account changes what the selective-interference effect can establish about memory architecture. Dual-representation accounts attribute the selective pattern to stronger disruption of sensory-bound representations associated with intrusive re-experiencing than of contextualized representations proposed to support coherent deliberate recollection (Brewin et al., 2010; Brewin, 2014). The retrieved-context model includes a change in stored associative structure: task events acquire new associations with film-associated context while the learned film associations remain available. Retrieval goals and test format then determine how strongly those new competitors affect performance. Preserved deliberate memory therefore does not determine whether the intervention selectively spared one film representation while disrupting another. Sensory-bound and contextualized representations may still contribute to trauma memory, but the selective-interference contrast alone does not identify their contribution.

The protection produced by the maintained film-category cue depends on an explicit modeling commitment. The implementation gives every film candidate additional support while task and filler candidates receive none. At the psychological level, this operation represents a maintained film-category cue of the kind implicated in category-directed recall; the simulations do not derive how participants construct that cue or which film features it contains. Task encoding strength has a similarly delimited role. Following established CMR parameterizations of encoding priority, it scales newly learned context-to-task associations (Polyn et al., 2009a; Sederberg et al., 2008; Talmi et al., 2019). It does not itself represent engagement, visuospatial processing, or a

particular intervention task. The simulations consequently establish what follows when competitor learning varies and film-directed support discriminates film from task material.

Interpreting Selective-Interference Evidence

The recognition analysis changes what the strongest positive selective-interference evidence can establish. Published demonstrations of reduced intrusion reports with preserved voluntary memory are dominated by designs that use recognition as the voluntary-memory test ([Asselbergs et al., 2024](#); [Holmes et al., 2009, 2010](#); [James et al., 2015](#); [Kessler et al., 2020](#); [Lau-Zhu et al., 2021](#)). In the model, a recognition probe does not need to defeat task candidates during context-to-item sampling because the candidate film item is already present. Preserved recognition can therefore accompany impaired context-guided film recall even when both tests depend on the same learned film associations. Such evidence establishes that presented film items can still recover useful contextual evidence, while leaving unresolved whether film material remains similarly available during deliberate recall.

Recall-based comparisons improve this inference but do not automatically isolate retrieval intentionality. Vigilance-intrusion tasks require film material to come to mind, yet they can present film cues and ask participants to monitor specifically for film intrusions ([Lau-Zhu et al., 2019, 2021](#)). External film cues can reinstate film-associated context, and film-monitoring instructions may encourage participants to maintain a film-oriented retrieval state. The present simulations include weak film cues during recall but do not activate a maintained film-category cue in the unguided condition. They therefore cannot determine whether vigilance-intrusion-task monitoring engages the same selective support used during deliberate recall. Mixed findings from more closely matched designs should motivate direct manipulations of cueing and retrieval goals rather than being explained retrospectively as consequences of those factors ([McConnell et al., 2026](#)).

Related procedures have been extended beyond analogue trauma films to real-life and occupational trauma, including motor vehicle accidents, emergency caesarean section, refugee trauma, and health-care work during COVID-19 ([Beckenstrom et al., 2026](#); [Horsch et al., 2017](#);

[Iyadurai et al., 2018, 2023](#); [Kanstrup et al., 2021, 2024](#); [Ramineni et al., 2023](#)). Broader reviews and replication attempts likewise caution that selective-interference effects are heterogeneous across trauma-film studies ([Asselbergs et al., 2023](#); [Hemi et al., 2023](#); [Varma et al., 2024](#); [Wessel et al., 2025](#)). The clinical reach of these procedures makes their mechanisms consequential, while the mixed evidence limits the conclusions that can be drawn from any single positive dissociation. Real-world extensions do not themselves establish that retrieved-context competition explains intervention effects outside the laboratory.

Intervention Tasks and Discriminating Tests

The account also suggests why intervention task content should matter. A task must become strongly enough associated with reminder-reinstated context to compete during later film retrieval. Selective preservation additionally requires task material to remain distinguishable from film material under the maintained film-category cue. Tetris can sustain substantial engagement, but active verbal comparison tasks can receive similar engagement ratings, and both visuospatial and verbal tasks have sometimes reduced intrusions ([Hagenaars et al., 2017](#); [Holmes et al., 2010](#); [Kessler et al., 2020](#)). Engagement may therefore enable competitor learning without explaining why deliberate film recall would exclude one task more successfully than another. The present model represents these proposed ingredients separately: task encoding strength controls competitor learning, whereas the maintained film-category cue assumes that film and task candidates can be discriminated.

A direct test would manipulate competitor strength and task-film discriminability independently. Participants could complete equally engaging post-reminder tasks whose contents vary in how readily a film-directed semantic or category cue distinguishes them from film material, followed by matched unguided and deliberate recall tests. Strong, discriminable task competitors should reduce unguided recall while leaving deliberate recall relatively preserved. Strong competitors that also receive support from the maintained film-category cue should impair both retrieval conditions, reducing or eliminating selective interference. Weak competitors should produce little interference regardless of their discriminability. Measuring task memory and

participants' retrieval strategies would test the assumptions that the current simulations impose. A parallel list-learning or category-recall experiment could evaluate the same prediction without depending on trauma-film-specific procedures.

Limitations and Scope

The simulations are qualitative mechanism tests rather than fits to selective-interference data. Core eCMR parameters were estimated from an independent affective free-recall dataset, but the trauma-film sequence and retrieval settings were introduced to examine the consequences of the proposed account. The simulations therefore demonstrate that retrieved-context competitor learning and discriminating retrieval support can produce the qualitative pattern. They do not estimate how often participants use these operations or establish that the operations generated any particular empirical result. A matched dataset containing externalized film and non-film retrieval sequences would provide a stronger basis for parameter anchoring and model comparison.

Several implementation choices further limit the account. Task encoding strength scales context-to-item learning without deriving that learning from task engagement or task content. The maintained film-category cue is perfectly selective, whereas real retrieval cues may support some competitors and fail to support some film details. The recognition simulation stops at item-to-context match evidence. Its film-source-matched foils are sufficient to test whether task items linked to film-associated context reduce the evidence separation between studied and unstudied film-source probes. They do not represent graded item similarity, and the analysis does not model false familiarity, false recollection, recollection rejection, confidence, or criterion placement in responses to similar lures ([Parks et al., 2026](#)). The recall simulations include test-phase film cues but do not represent all monitoring demands of diary or vigilance-intrusion procedures. Finally, the model addresses episodic retrieval processes relevant to intrusive remembering; it is not a model of PTSD, clinical outcome, or treatment efficacy.

The broader contribution is to connect an account of how competitors become bound to target-associated context with an account of how goal-directed retrieval remains selective. Context can continue to provide access to an event while losing its ability to distinguish that event

from later material. Deliberate recall is protected when a retrieval goal organizes search around information that separates targets from those competitors, whereas item-presenting tests change the route by which the target enters retrieval. Selective interference in the trauma-film paradigm provides a consequential case for specifying these interactions. Grounding that case in general memory theory also sharpens the evidence needed to identify memory processes that are distinctive to trauma.

References

- Asselbergs, J., Riper, H., Engelhard, I. M., Mannes, F., & Sijbrandij, M. (2024). The effectiveness of two novel approaches to prevent intrusions: A pilot study comparing Tetris_dualtask and imagery rescripting to control. *Journal of Behavior Therapy and Experimental Psychiatry*, 82, 101920. <https://doi.org/10.1016/j.jbtep.2023.101920>
- Asselbergs, J., van Bentum, J., Riper, H., Cuijpers, P., Holmes, E. A., & Sijbrandij, M. (2023). A systematic review and meta-analysis of the effect of cognitive interventions to prevent intrusive memories using the trauma film paradigm. *Journal of Psychiatric Research*, 159, 116–129. <https://doi.org/10.1016/j.jpsychires.2023.01.028>
- Baddeley, A. D., & Andrade, J. (2000). Working memory and the vividness of imagery. *Journal of Experimental Psychology: General*, 129(1), 126–145. <https://doi.org/10.1037/0096-3445.129.1.126>
- Beckenstrom, A. C., Bonsall, M. B., Markham, A., Slade, O., Ramineni, V., Iyadurai, L., Islam, Z., Highfield, J., Jaki, T., Goodwin, G. M., Dias, R., Daniels, R., Malik, A., Summers, C., Kingslake, J., & Holmes, E. A. (2026). A digital imagery-competing task intervention for stopping intrusive memories in trauma-exposed health-care staff during the COVID-19 pandemic in the UK: A bayesian adaptive randomised clinical trial. *The Lancet Psychiatry*, 13(3), 233–247. [https://doi.org/10.1016/s2215-0366\(25\)00397-9](https://doi.org/10.1016/s2215-0366(25)00397-9)
- Bird, C. M., Bisby, J. A., & Burgess, N. (2012). The hippocampus and spatial constraints on mental imagery. *Frontiers in Human Neuroscience*, 6, 142. <https://doi.org/10.3389/fnhum.2012.00142>

- Bisby, J. A., Burgess, N., & Brewin, C. R. (2020). Reduced memory coherence for negative events and its relationship to posttraumatic stress disorder. *Current Directions in Psychological Science*, 29(3), 267–272. <https://doi.org/10.1177/0963721420917691>
- Brewin, C. R. (2014). Episodic memory, perceptual memory, and their interaction: Foundations for a theory of posttraumatic stress disorder. *Psychological Bulletin*, 140(1), 69–97. <https://doi.org/10.1037/a0033722>
- Brewin, C. R., Dalgleish, T., & Joseph, S. (1996). A dual representation theory of posttraumatic stress disorder. *Psychological Review*, 103(4), 670–686. <https://doi.org/10.1037/0033-295X.103.4.670>
- Brewin, C. R., Gregory, J. D., Lipton, M., & Burgess, N. (2010). Intrusive images in psychological disorders: Characteristics, neural mechanisms, and treatment implications. *Psychological Review*, 117(1), 210–232. <https://doi.org/10.1037/a0018113>
- Bussemeyer, J. R., & Wang, Y.-M. (2000). Model comparisons and model selections based on generalization criterion methodology. *Journal of Mathematical Psychology*, 44(1), 171–189. <https://doi.org/10.1006/jmps.1999.1282>
- Cohen, R. T., & Kahana, M. J. (2022). A memory-based theory of emotional disorders. *Psychological Review*, 129(4), 742–776. <https://doi.org/10.1037/rev0000334>
- Deepröse, C., Zhang, S., DeJong, H., Dalgleish, T., & Holmes, E. A. (2012). Imagery in the aftermath of viewing a traumatic film: Using cognitive tasks to modulate the development of involuntary memory. *Journal of Behavior Therapy and Experimental Psychiatry*, 43(2), 758–764. <https://doi.org/10.1016/j.jbtep.2011.10.008>
- Dupertuys, J., Shulhan, A., & Talmi, D. (2026). *Ukraine Memory experiment data set*.
- Ehlers, A., & Clark, D. M. (2000). A cognitive model of posttraumatic stress disorder. *Behaviour Research and Therapy*, 38(4), 319–345. [https://doi.org/10.1016/S0005-7967\(99\)00123-0](https://doi.org/10.1016/S0005-7967(99)00123-0)
- Hagenaars, M. A., Holmes, E. A., Klaassen, F., & Elzinga, B. (2017). Tetris and word games lead to fewer intrusive memories when applied several days after analogue trauma. *European Journal of Psychotraumatology*, 8(sup1), 1386959.

<https://doi.org/10.1080/20008198.2017.1386959>

Healey, M. K., & Kahana, M. J. (2014). Is memory search governed by universal principles or idiosyncratic strategies? *Journal of Experimental Psychology: General*, 143(2), 575–596.

<https://doi.org/10.1037/a0033715>

Healey, M. K., & Kahana, M. J. (2016). A four-component model of age-related memory change. *Psychological Review*, 123(1), 23–69. <https://doi.org/10.1037/rev0000015>

Healey, M. K., & Wahlheim, C. N. (2024). PEPPR: A post-encoding pre-production reinstatement model of dual-list free recall. *Memory & Cognition*, 52(1), 163–181.

<https://doi.org/10.3758/s13421-023-01453-z>

Hemi, A., Sopp, M. R., Perel, A., Holmes, E. A., & Levy-Gigi, E. (2023). Cognitive flexibility moderates the efficacy of a visuospatial intervention following exposure to analog trauma. *Journal of Behavior Therapy and Experimental Psychiatry*, 81, 101858.

<https://doi.org/10.1016/j.jbtep.2023.101858>

Herrema, H. G., & Kahana, M. J. (2024). *First recall costs and benefits*. PsyArXiv.

<https://doi.org/10.31234/osf.io/sm3fw>

Holmes, E. A., & Bourne, C. (2008). Inducing and modulating intrusive emotional memories: A review of the trauma film paradigm. *Acta Psychologica*, 127(3), 553–566.

<https://doi.org/10.1016/j.actpsy.2007.11.002>

Holmes, E. A., Brewin, C. R., & Hennessey, R. G. (2004). Trauma films, information processing, and intrusive memory development. *Journal of Experimental Psychology: General*, 133(1), 3–22. <https://doi.org/10.1037/0096-3445.133.1.3>

Holmes, E. A., James, E. L., Coode-Bate, T., & Deeproose, C. (2009). Can playing the computer game Tetris reduce the build-up of flashbacks for trauma? A proposal from cognitive science. *PLoS ONE*, 4(1), e4153. <https://doi.org/10.1371/journal.pone.0004153>

Holmes, E. A., James, E. L., Kilford, E. J., & Deeproose, C. (2010). Key steps in developing a cognitive vaccine against traumatic flashbacks: Visuospatial Tetris versus verbal Pub Quiz. *PLoS ONE*, 5(11), e13706. <https://doi.org/10.1371/journal.pone.0013706>

- Horsch, A., Vial, Y., Favrod, C., Harari, M. M., Blackwell, S. E., Watson, P., Iyadurai, L., Bonsall, M. B., & Holmes, E. A. (2017). Reducing intrusive traumatic memories after emergency caesarean section: A proof-of-principle randomized controlled study. *Behaviour Research and Therapy*, 94, 36–47. <https://doi.org/10.1016/j.brat.2017.03.018>
- Howard, M. W., & Kahana, M. J. (1999). Contextual variability and serial position effects in free recall. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 25(4), 923–941. <https://doi.org/10.1037/0278-7393.25.4.923>
- Howard, M. W., & Kahana, M. J. (2002). A distributed representation of temporal context. *Journal of Mathematical Psychology*, 46(3), 269–299. <https://doi.org/10.1006/jmps.2001.1388>
- Howard, M. W., Youker, T. E., & Venkatadass, V. S. (2008). The persistence of memory: Contiguity effects across hundreds of seconds. *Psychonomic Bulletin & Review*, 15(1), 58–63.
- Hupbach, A., Hardt, O., Gomez, R., & Nadel, L. (2008). The dynamics of memory: Context-dependent updating. *Learning & Memory*, 15(8), 574–579. <https://doi.org/10.1101/lm.1022308>
- Iyadurai, L., Blackwell, S. E., Meiser-Stedman, R., Watson, P. C., Bonsall, M. B., Geddes, J. R., Nobre, A. C., & Holmes, E. A. (2018). Preventing intrusive memories after trauma via a brief intervention involving Tetris computer game play in the emergency department: A proof-of-concept randomized controlled trial. *Molecular Psychiatry*, 23(3), 674–682. <https://doi.org/10.1038/mp.2017.23>
- Iyadurai, L., Highfield, J., Kanstrup, M., Markham, A., Ramineni, V., Guo, B., Jaki, T., Kingslake, J., Goodwin, G. M., Summers, C., Bonsall, M. B., & Holmes, E. A. (2023). Reducing intrusive memories after trauma via an imagery-competing task intervention in COVID-19 intensive care staff: A randomised controlled trial. *Translational Psychiatry*, 13(1), 290. <https://doi.org/10.1038/s41398-023-02578-0>
- Iyadurai, L., Visser, R. M., Lau-Zhu, A., Porcheret, K., Horsch, A., Holmes, E. A., & James, E. L. (2019). Intrusive memories of trauma: A target for research bridging cognitive science and its

clinical application. *Clinical Psychology Review*, 69, 67–82.

<https://doi.org/10.1016/j.cpr.2018.08.005>

James, E. L., Bonsall, M. B., Hoppitt, L., Tunbridge, E. M., Geddes, J. R., Milton, A. L., & Holmes, E. A. (2015). Computer game play reduces intrusive memories of experimental trauma via reconsolidation-update mechanisms. *Psychological Science*, 26(8), 1201–1215.

<https://doi.org/10.1177/0956797615583071>

James, E. L., Lau-Zhu, A., Clark, I. A., Visser, R. M., Hageraars, M. A., & Holmes, E. A. (2016). The trauma film paradigm as an experimental psychopathology model of psychological trauma: Intrusive memories and beyond. *Clinical Psychology Review*, 47, 106–142.

<https://doi.org/10.1016/j.cpr.2016.04.010>

Jin, B. J., Halpern, D. J., & Kahana, M. J. (2024). *A theory of memory for items and associations*.

<https://memory.psych.upenn.edu/files/pubs/JinEtal24.pdf>

Kahana, M. J. (1996). Associative retrieval processes in free recall. *Memory & Cognition*, 24(1), 103–109.

Kahana, M. J. (2020). Computational models of memory search. *Annual Review of Psychology*, 71, 107–138. <https://doi.org/10.1146/annurev-psych-010418-103358>

Kanstrup, M., Kontio, E., Geranmayeh, A., Olofsdotter Lauri, K., Moulds, M. L., & Holmes, E. A. (2021). A single case series using visuospatial task interference to reduce the number of visual intrusive memories of trauma with refugees. *Clinical Psychology & Psychotherapy*, 28(1), 109–123. <https://doi.org/10.1002/cpp.2489>

Kanstrup, M., Singh, L., Leehr, E. J., Göransson, K. E., Pihlgren, S. A., Iyadurai, L., Dahl, O., Falk, A.-C., Lindström, V., Hadziosmanovic, N., Gabrysch, K., Moulds, M. L., & Holmes, E. A. (2024). A guided single session intervention to reduce intrusive memories of work-related trauma: A randomised controlled trial with healthcare workers in the COVID-19 pandemic. *BMC Medicine*, 22(1), 403. <https://doi.org/10.1186/s12916-024-03569-8>

Kessler, H., Schmidt, A.-C., James, E. L., Blackwell, S. E., von Rauchhaupt, M., Harren, K., Kehyayan, A., Clark, I. A., Sauvage, M., Herpertz, S., Axmacher, N., & Holmes, E. A. (2020).

Visuospatial computer game play after memory reminder delivered three days after a traumatic film reduces the number of intrusive memories of the experimental trauma. *Journal of Behavior Therapy and Experimental Psychiatry*, 67, 101454.

<https://doi.org/10.1016/j.jbtep.2019.01.006>

Kindt, M., Soeter, M., & Vervliet, B. (2009). Beyond extinction: Erasing human fear responses and preventing the return of fear. *Nature Neuroscience*, 12(3), 256–258.

<https://doi.org/10.1038/nn.2271>

Kragel, J. E., Morton, N. W., & Polyn, S. M. (2015). Neural activity in the medial temporal lobe reveals the fidelity of mental time travel. *Journal of Neuroscience*, 35(7), 2914–2926.

<https://doi.org/10.1523/JNEUROSCI.3378-14.2015>

Krans, J., Näring, G., Becker, E. S., & Holmes, E. A. (2009). Intrusive trauma memory: A review and functional analysis. *Applied Cognitive Psychology*, 23(8), 1076–1088.

<https://doi.org/10.1002/acp.1611>

Lau-Zhu, A., Henson, R. N., & Holmes, E. A. (2019). Intrusive memories and voluntary memory of a trauma film: Differential effects of a cognitive interference task after encoding. *Journal of Experimental Psychology: General*, 148(12), 2154–2180. <https://doi.org/10.1037/xge0000598>

Lau-Zhu, A., Henson, R. N., & Holmes, E. A. (2021). Selectively interfering with intrusive but not voluntary memories of a trauma film: Accounting for the role of associative memory. *Clinical Psychological Science*, 9(6), 1128–1143. <https://doi.org/10.1177/2167702621998315>

Lohnas, L. J., Healey, M. K., & Davachi, L. (2023). Neural temporal context reinstatement of event structure during memory recall. *Journal of Experimental Psychology: General*, 152(7), 1840–1872. <https://doi.org/10.1037/xge0001354>

Lohnas, L. J., Polyn, S. M., & Kahana, M. J. (2015). Expanding the scope of memory search: Modeling intralist and interlist effects in free recall. *Psychological Review*, 122(2), 337–363. <https://doi.org/10.1037/a0039036>

Lohnas, L. J., Polyn, S. M., & Kahana, M. J. (2026). Specialized recall procedures. *Psychonomic Bulletin & Review*, 33(3), 96. <https://doi.org/10.3758/s13423-025-02744-z>

- Luce, R. D. (1959). *Individual choice behavior: A theoretical analysis*. Wiley.
- McConnell, T. G. W., Bernhard, H., Holmes, E. A., Henson, R., & Talmi, D. (2026). *Testing the dual representation account of memory for trauma*. PsyArXiv preprint.
https://osf.io/preprints/psyarxiv/m3pnz_v1/
- Moccia, A., & Morcom, A. M. (2022). Cue overlap supports preretrieval selection in episodic memory: ERP evidence. *Cognitive, Affective, & Behavioral Neuroscience*, 22, 492–508.
<https://doi.org/10.3758/s13415-021-00971-0>
- Moccia, A., Plummer, M., Simpson, I., & Morcom, A. M. (2025). The effects of external cue overlap and internal goals on selective memory retrieval as revealed by electroencephalographic (EEG) neural pattern reinstatement. *European Journal of Neuroscience*, 62(1), e70194. <https://doi.org/10.1111/ejn.70194>
- Morton, N. W., & Polyn, S. M. (2016). A predictive framework for evaluating models of semantic organization in free recall. *Journal of Memory and Language*, 86, 119–140.
<https://doi.org/10.1016/j.jml.2015.10.002>
- Norman, Y., Yeagle, E. M., Harel, M., Mehta, A. D., & Malach, R. (2017). Neuronal baseline shifts underlying boundary setting during free recall. *Nature Communications*, 8(1), 1301.
<https://doi.org/10.1038/s41467-017-01184-1>
- Parks, C. M., Yonelinas, A. P., & Wahlheim, C. N. (2026). Towards a unified theory of false memory for similar episodes. *Psychonomic Bulletin & Review*, 33, 88.
<https://doi.org/10.3758/s13423-025-02855-7>
- Polyn, S. M., Erlikhman, G., & Kahana, M. J. (2011). Semantic cuing and the scale insensitivity of recency and contiguity. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 37(3), 766–775. <https://doi.org/10.1037/a0022475>
- Polyn, S. M., Kragel, J. E., Morton, N. W., McCluey, J. D., & Cohen, Z. D. (2012). The neural dynamics of task context in free recall. *Neuropsychologia*, 50(4), 447–457.
<https://doi.org/10.1016/j.neuropsychologia.2011.08.025>
- Polyn, S. M., Norman, K. A., & Kahana, M. J. (2009a). A context maintenance and retrieval

- model of organizational processes in free recall. *Psychological Review*, 116(1), 129–156.
<https://doi.org/10.1037/a0014420>
- Polyn, S. M., Norman, K. A., & Kahana, M. J. (2009b). Task context and organization in free recall. *Neuropsychologia*, 47(11), 2158–2163.
<https://doi.org/10.1016/j.neuropsychologia.2009.02.013>
- Raaijmakers, J. G. W., & Shiffrin, R. M. (1980). SAM: A theory of probabilistic search of associative memory. In G. H. Bower (Ed.), *The psychology of learning and motivation: Advances in research and theory* (Vol. 14, pp. 207–262). Academic Press.
[https://doi.org/10.1016/S0079-7421\(08\)60162-0](https://doi.org/10.1016/S0079-7421(08)60162-0)
- Ramineni, V., Millroth, P., Iyadurai, L., Jaki, T., Kingslake, J., Highfield, J., Summers, C., Bonsall, M. B., & Holmes, E. A. (2023). Treating intrusive memories after trauma in healthcare workers: A bayesian adaptive randomised trial developing an imagery-competing task intervention. *Molecular Psychiatry*, 28(7), 2985–2994.
<https://doi.org/10.1038/s41380-023-02062-7>
- Schwartz, G., Howard, M. W., Jing, B., & Kahana, M. J. (2005). Shadows of the past: Temporal retrieval effects in recognition memory. *Psychological Science*, 16(11), 898–904.
<https://doi.org/10.1111/j.1467-9280.2005.01634.x>
- Sederberg, P. B., Gershman, S. J., Polyn, S. M., & Norman, K. A. (2011). Human memory reconsolidation can be explained using the temporal context model. *Psychonomic Bulletin & Review*, 18(3), 455–468. <https://doi.org/10.3758/s13423-011-0086-9>
- Sederberg, P. B., Howard, M. W., & Kahana, M. J. (2008). A context-based theory of recency and contiguity in free recall. *Psychological Review*, 115(4), 893–912.
<https://doi.org/10.1037/a0013396>
- Talmi, D. (2013). Enhanced emotional memory: Cognitive and neural mechanisms. *Current Directions in Psychological Science*, 22(6), 430–436.
<https://doi.org/10.1177/0963721413498893>
- Talmi, D., Lohnas, L. J., & Daw, N. D. (2019). A retrieved context model of the emotional

modulation of memory. *Psychological Review*, 126(4), 455–485.

<https://doi.org/10.1037/rev0000132>

Talmi, D., Luk, B. T. C., McGarry, L. M., & Moscovitch, M. (2007). The contribution of relatedness and distinctiveness to emotionally-enhanced memory. *Journal of Memory and Language*, 56(4), 555–574. <https://doi.org/10.1016/j.jml.2007.01.002>

Talmi, D., & Moscovitch, M. (2004). Can semantic relatedness explain the enhancement of memory for emotional words? *Memory & Cognition*, 32(5), 742–751.

<https://doi.org/10.3758/bf03195864>

Unsworth, N., Brewer, G. A., & Spillers, G. J. (2010). Understanding the dynamics of correct and error responses in free recall: Evidence from externalized free recall. *Memory & Cognition*, 38(4), 419–430. <https://doi.org/10.3758/MC.38.4.419>

Varma, M. M., Zeng, S., Singh, L., Holmes, E. A., Huang, J., Chiu, M. H., & Hu, X. (2024). A systematic review and meta-analysis of experimental methods for modulating intrusive memories following lab-analogue trauma exposure in non-clinical populations. *Nature Human Behaviour*, 8(10), 1968–1987. <https://doi.org/10.1038/s41562-024-01956-y>

Wessel, I., Krans, J., Albers, C. J., Chauhan, N., Ferreira de Sá, D. S., Hauck, A., Jaswetz, L., Michael, T., Nixon, R. D. V., Pearson, D. G., Slattery, I., Smeets, T., Takarangi, M. K. T., Willems, T. R. A., & van Schie, K. (2025). Evidence that Tetris reduces immediate but not subsequent daily intrusions of a trauma film: A multilab replication study. *Collabra: Psychology*, 11(1), 130791. <https://doi.org/10.1525/collabra.130791>

Zaromb, F. M., Howard, M. W., Dolan, E. D., Sirotin, Y. B., Tully, M., Wingfield, A., & Kahana, M. J. (2006). Temporal associations and prior-list intrusions in free recall. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 32(4), 792–804.

<https://doi.org/10.1037/0278-7393.32.4.792>

Zhou, C. Y., Talmi, D., Daw, N. D., & Mattar, M. G. (2025). Episodic retrieval for model-based evaluation in sequential decision tasks. *Psychological Review*, 132(1), 18–49.

<https://doi.org/10.1037/rev0000505>

Table 1

Core notation used in the model specification. Parameters are defined with the operations they control, and numerical values are reported with the simulation methodology.

Symbol	Role	Main use
f_i	Item feature vector for encoded event i	Encoding, reminders, recall updates, recognition probes
s_i	Source feature vector distinguishing film from non-film items	Source-context encoding
c_i	Context state after encoded event i	Encoding and learning
M^{FC}	Feature-to-context associations	Context reinstatement from encoded, cued, sampled, or probed items
M^{CF}	Context-to-feature associations	Candidate support during context-guided recall
α	Background support from nonpaired item-associated context units	Initial M^{CF}
δ	Support from each item's paired context unit	Initial M^{CF}
β_{enc}	Drift toward item-cued context input during encoding	Film, task, and filler encoding
β_{rec}	Drift toward a sampled item's associated context	Recall updates
β_{emot}	Drift in the source-context component	Source-context encoding
γ	Feature-to-context learning rate	Updates to M^{FC}
λ	Context-to-feature learning	Updates to M^{CF}

Table 2

Default numerical parameter values. Values are from a pooled eCMR fit to the immediate free-recall sequences from Dupertuys et al. (2026), with α fixed to zero, except for the source-context and primacy settings described in the text.

Symbol	Value	Role
β_{enc}	0.440	Drift toward item-cued context during encoding
β_{start}	0.766	Drift toward the start-of-film context during start-of-film reinstatement
β_{rec}	0.678	Drift toward sampled-item context during recall
γ	0.468	Feature-to-context learning
δ	0.00022	Initial support from each item's paired context unit
α	0	Initial background support from other item-associated context units
ϕ_s	0.100	Scale parameter for the primacy profile ϕ_i
ϕ_d	1.500	Decay parameter for the primacy profile ϕ_i
τ	3.562	Luce-choice exponent for candidate sampling
β_{emot}	0.696	Drift toward the active source feature
ϕ_{emot}	0.130	Added source-context learning for film items
η_s	0.050	Baseline source-context-to-feature